

3 TIMED TESTS · 50 QUESTIONS EACH

# Gastrointestinal

*Practice tests of USMLE-style questions drawn from the website quizzes, with answers and explanations.*

For educational use only. This is a study aid, not a clinical reference.

# Instructions



**60 minutes**
for each test

**50 questions**
in each test

**About 72 seconds**
for each question

**Every question comes from the website.** These tests are built from the quizzes on the Clinical Conditions website, 150 of its 462 questions, so if you have done the website quizzes you will meet questions you have seen before. For the truest measure, take a test before using the website quiz for these topics, and treat a remembered answer as revision rather than a sign of mastery.

## TAKING A TEST

- Take each test in one sitting, with a timer set to 60 minutes. Do not pause the timer.
- Each question has five options and one best answer. Choose the single best option, even if more than one seems reasonable.
- Fill in the answer sheet after the last question of each test, or use a separate piece of paper.
- Do not use notes, the website, or other references. The abbreviation list on the next page is allowed.
- There is no penalty for guessing, so answer every question. An unanswered question counts as incorrect.
- When time ends, stop, even if you have not finished.

## ABOUT THE TESTS

- No question appears in more than one test, so the three tests can be taken in any order.
- Every test has 15 physiology and 35 clinical questions, covering all 7 physiology topics and all 30 clinical topics, with a similar mix of Step 1 questions (mechanisms, microbiology, and pathology) and Step 2 questions (diagnosis, next step, and management).
- Questions are in random order, and the topic being tested is not shown until the answers.

## SCORING AND REVIEW

- Your score is the number of correct answers. The answer key at the start of each answer section has the same layout as the answer sheet, so you can mark one against the other.
- Then read the explanations. Each one names the topic and step, so you can see what to revisit.
- Answers follow the course material. Where a guideline differs, the explanation says so under "Additional information".

## CONTENTS

| | | | | | |
|---|---|---|---|---|---|
| Test A | **Questions** | Test B | **Questions** | Test C | **Questions** |
| Test A | **Answer sheet** | Test B | **Answer sheet** | Test C | **Answer sheet** |
| Test A | **Answers** | Test B | **Answers** | Test C | **Answers** |

# Abbreviations

ALLOWED DURING A TEST

**5-HT3** Serotonin receptor 3, an ion channel

**5-HT4** Serotonin receptor 4

**ABX** Antibiotics

**ACh** Acetylcholine

**AFP** Alpha-fetoprotein

**ALP** Alkaline phosphatase

**ALT** Alanine aminotransferase

**AMA** Anti-mitochondrial antibody

**anti-CCP** Anti-cyclic citrullinated peptide antibody

**anti-dsDNA** Anti-double-stranded DNA antibody

**anti-HBc** Antibody to hepatitis B core antigen

**anti-HDV** Antibody to hepatitis D virus

**anti-tTG** Anti-tissue transglutaminase antibody

**AQP4** Aquaporin-4

**AST** Aspartate aminotransferase

**ATP7A** Copper-transporting ATPase A

**ATP7B** Copper-transporting ATPase B

**BUN** Blood urea nitrogen

**cAMP** Cyclic adenosine monophosphate

**cAMP** Cyclic adenosine monophosphate

**CCK** Cholecystokinin: released by the duodenum in response to fat; makes the gallbladder contract

**CD4** Cluster of differentiation 4

**CK** Creatine kinase

**CN** Cranial nerve

**CN X** Cranial nerve X (the vagus nerve)

**COPD** Chronic obstructive pulmonary disease

**CRP** C-reactive protein

**CSF** Cerebrospinal fluid

**CT** Computed tomography

**CT AP** Computed tomography of the abdomen and pelvis

**CTR1** Copper transporter 1

**CXR** Chest x-ray

**CYP2E1** Cytochrome P450 2E1

**CYP7A1** Cholesterol 7α-hydroxylase

**D2** Dopamine receptor 2

**DMT1** Divalent metal transporter 1

**DVT** Deep vein thrombosis

**ECL** Enterochromaffin-like

**ERCP** Endoscopic retrograde cholangiopancreatography

**FAST** Focused assessment with sonography in trauma

**FBC** Full blood count

**FODMAP** Fermentable oligosaccharides, disaccharides, monosaccharides, and polyols

**GE** Gastroesophageal

**GERD** Gastroesophageal reflux disease

**GGT** Gamma-glutamyl transferase

**GIP** Glucose-dependent insulinotropic peptide

**GLUT2** Glucose transporter 2

**H1** Histamine receptor 1

**H2** Histamine receptor 2

**H2R** Histamine H2 receptor

**HBIG** Hepatitis B immune globulin

**HBsAg** Hepatitis B surface antigen: current infection

**HBV** Hepatitis B virus

**HCl** Hydrochloric acid

**HCV** Hepatitis C virus

**HDV** Hepatitis D (delta) virus

**HELLP** Hemolysis, elevated liver enzymes, low platelets

**HFE** Hereditary hemochromatosis gene

**HIDA** Hepatobiliary iminodiacetic acid scan (cholescintigraphy)

**HIV** Human immunodeficiency virus

**HIV** Human immunodeficiency virus

**HLA** Human leukocyte antigen

**IBD** Inflammatory bowel disease

**IBS** Irritable bowel syndrome

**ICU** Intensive care unit

**IgA** Immunoglobulin A

**IgE** Immunoglobulin E

**IgG** Immunoglobulin G

**IgG4** Immunoglobulin G4

**IgM** Immunoglobulin M

**IM** Intramuscular

**IM** Intramuscular

**INR** International normalized ratio

**IV** Intravenous

**KUB** Kidneys, ureters, and bladder

**LCHAD** Long-chain 3-hydroxyacyl-CoA dehydrogenase, an enzyme of fatty acid beta-oxidation

**LDH** Lactate dehydrogenase

**LDL** Low-density lipoprotein

**LDL** Low-density lipoprotein

**LES** Lower esophageal sphincter

**LFTs** Liver function tests

**M3** Muscarinic acetylcholine receptor 3

**MALT** Mucosa-associated lymphoid tissue

**MASLD** Metabolic dysfunction-associated steatotic liver disease (formerly NAFLD)

**MCAD** Medium-chain acyl-CoA dehydrogenase

**MCV** Mean corpuscular volume

**MRCP** Magnetic resonance cholangiopancreatography

**MRI** Magnetic resonance imaging

**MRP2** Multidrug resistance-associated protein 2

**NAPQI** N-acetyl-p-benzoquinone imine

**NSAID** Nonsteroidal anti-inflammatory drug

**NSAID** Nonsteroidal anti-inflammatory drug

**NSAIDs** Nonsteroidal anti-inflammatory drugs

**p-ANCA** Perinuclear anti-neutrophil cytoplasmic antibody

**PAS** Periodic acid-Schiff

**PBC** Primary biliary cholangitis (formerly primary biliary cirrhosis)

**PCR** Polymerase chain reaction

**PepT1** Peptide transporter 1

**PET** Positron emission tomography

**PG** Prostaglandins

**pIgR** Polymeric immunoglobulin receptor

**PPI** Proton pump inhibitor

**PPIs** Proton pump inhibitors

**PYY** Peptide YY

**RLQ** Right lower quadrant

**RNA** Ribonucleic acid

**SGLT1** Sodium-glucose cotransporter 1

**SMA** Superior mesenteric artery

**TIPS** Transjugular intrahepatic portosystemic shunt

**TPMT** Thiopurine methyltransferase

**TPN** Total parenteral nutrition

**UGT** UDP-glucuronosyltransferase

**WBC** White blood cells

# Test A

50 questions | 60 minutes | Start the timer

**1** An 81-year-old woman with long-standing gallstones has chronic watery diarrhea and weight loss. Imaging shows air in the biliary tree and an abnormal connection between the gallbladder and the hepatic flexure of the colon.

**Which best explains her diarrhea?**

(A) Lactose intolerance
(B) Pancreatic enzyme deficiency
(C) Bacterial overgrowth in the stomach
(D) Bile salts entering the colon directly
(E) Small bowel obstruction

**2** A 56-year-old woman with primary biliary cholangitis has fingers that turn white in the cold, tight skin over her fingers, small dilated blood vessels on her face, and difficulty swallowing.

**Which antibody is most likely positive for her second condition?**

(A) Anti-Jo-1 antibody
(B) Anti-centromere antibody
(C) Anti-CCP antibody
(D) Anti-Scl-70 (topoisomerase I) antibody
(E) Anti-dsDNA antibody

**3** A 58-year-old man with cirrhosis has breast enlargement, small testes, reddened palms, and many small, spider-like blood vessels on his chest.

**Which mechanism best explains these findings?**

(A) Decreased albumin synthesis by the failing liver
(B) Increased estrogen from reduced hepatic clearance
(C) Vitamin K deficiency from reduced bile salt secretion
(D) Increased ammonia from reduced hepatic clearance
(E) Portal hypertension

**4** A pharmacology student asks why the area postrema can respond to toxins in the blood when most of the brainstem cannot.

**Which feature of the area postrema explains this?**

(A) Its ependyma actively pumps toxins in from the CSF
(B) It lies in the floor of the third ventricle, beside the hypothalamus
(C) It is bathed in lymph drained from the abdominal organs
(D) Its capillaries lack tight junctions, so it sits outside the blood-brain barrier
(E) Its neurons send axons out through the vagus nerve to the gut

**5** A 24-year-old man with hemolytic anemia has jaundice. His urine and stool are normal in color.

**Which type of hyperbilirubinemia is most likely?**

(A) Unconjugated, as it is bound to albumin
(B) Unconjugated, as it is water soluble
(C) Conjugated, as MRP2 is defective
(D) Conjugated, as bile is obstructed
(E) Conjugated, as it is water soluble

**6** A 66-year-old man with rheumatoid arthritis needs long-term NSAID therapy and has a history of a gastric ulcer. He is started on a drug that is a prostaglandin E1 analog to protect the gastric mucosa.

**This drug is contraindicated in which group of patients?**

(A) Pregnant patients
(B) Patients with asthma
(C) Patients with renal impairment
(D) Patients taking warfarin
(E) Patients with hypothyroidism

**7** A 29-year-old man with moderate-to-severe Crohn disease that has not responded to corticosteroids and azathioprine is to start infliximab.

**Which test should be performed before starting therapy?**

- (A) Serum ceruloplasmin
- (B) Thiopurine methyltransferase (TPMT) enzyme activity level
- (C) Interferon-gamma release assay (or tuberculin skin test)
- (D) Pulmonary function tests
- (E) Anti-Saccharomyces cerevisiae antibodies

**8** A 48-year-old woman has 18 hours of constant right upper quadrant pain radiating to her right shoulder, with nausea. Temperature is 38.4°C. She stops breathing in when the right upper quadrant is palpated. WBC is 15800/µL; bilirubin, ALP, and lipase are normal. Ultrasound shows gallstones, a 5-mm gallbladder wall, and fluid around the gallbladder.

**Which is the most appropriate definitive management?**

- (A) IV ABX and laparoscopic cholecystectomy during this admission
- (B) Oral ursodeoxycholic acid
- (C) ERCP
- (D) Outpatient follow-up in 6 weeks
- (E) Percutaneous cholecystostomy as first-line in all patients

**9** A 25-year-old woman eats table sugar. Enzymes on the brush border of her enterocytes digest it before it can be absorbed.

**Which enzyme acts, and what does it form?**

- (A) Sucrase, forming glucose and galactose
- (B) Sucrase, forming glucose and fructose
- (C) Lactase, forming glucose and galactose
- (D) Lactase, forming glucose alone
- (E) Maltase, forming glucose and fructose

**10** A 50-year-old woman with a duodenal ulcer and confirmed H. pylori infection needs a drug that coats the ulcer base and also kills the organism.

**Which drug is most appropriate?**

- (A) Aluminum hydroxide gel
- (B) Sucralfate
- (C) Bismuth subcitrate
- (D) Famotidine
- (E) Misoprostol

**11** A 26-year-old woman at 30 weeks' gestation develops fever, nausea, and jaundice 5 weeks after returning from a trip where she drank untreated water. Within days she becomes confused, with an INR of 2.8.

**Which virus most likely caused her illness?**

- (A) Hepatitis D virus
- (B) Hepatitis B virus
- (C) Hepatitis A virus
- (D) Hepatitis E virus
- (E) Hepatitis C virus

**12** A 26-year-old woman develops watery diarrhea 3 days after arriving in Mexico. She has 4 loose stools a day, no fever and no blood in the stool, and is drinking well.

**Which is the most appropriate management?**

- (A) Oral rehydration, with loperamide if needed
- (B) Stool ova and parasites, then empirical metronidazole
- (C) IV ciprofloxacin and admission
- (D) Oral vancomycin
- (E) Urgent colonoscopy

**13** A 2-week-old infant suddenly vomits green fluid and becomes unwell. Imaging suggests abnormal positioning of the bowel with a twisted mesentery.

**Which developmental step failed?**

- (A) Closure of the abdominal wall
- (B) Fusion of the pancreatic ducts
- (C) Recanalization of the solid duodenum after week 8
- (D) Separation of trachea and esophagus
- (E) The 270° rotation of the midgut around the SMA

**14** A 48-year-old man is diagnosed with HFE hereditary hemochromatosis. His hemoglobin is 15.2 g/dL, ferritin is 1600 ng/mL, and he has no cardiac disease.

**Which is the most appropriate treatment?**

(A) Low-iron diet alone
(B) Liver transplant
(C) D-penicillamine
(D) Oral deferasirox
(E) Therapeutic phlebotomy

**15** A 38-year-old woman with poorly controlled type 2 diabetes develops severe epigastric pain radiating to her back. Her blood sample looks milky, and lipase is markedly raised. She does not drink alcohol, and ultrasound shows no gallstones.

**Which laboratory finding most likely explains her pancreatitis?**

(A) Serum triglycerides of 2400 mg/dL
(B) Elevated serum ferritin
(C) Positive anti-mitochondrial antibody
(D) Serum calcium of 8.0 mg/dL
(E) Elevated serum IgG4

**16** A 64-year-old woman with autoimmune gastritis has megaloblastic anemia, low serum B12, and reduced vibration sense in her feet.

**Which is the most appropriate treatment?**

(A) Oral folic acid alone
(B) Intramuscular vitamin B12
(C) Oral iron
(D) Proton pump inhibitor
(E) H. pylori eradication therapy

**17** A 49-year-old woman with PBC starts ursodeoxycholic acid.

**How does this drug help her?**

(A) It displaces toxic bile salts and increases bile flow
(B) It blocks viral reverse transcriptase in the hepatocyte
(C) It binds bile salts in the gut and blocks their reuptake
(D) It inhibits CYP7A1, stopping synthesis of new bile salts
(E) It suppresses T cells that are attacking her bile ducts

**18** A 48-year-old man is given a somatostatin analog during a research study of gastric function.

**Which cells does somatostatin inhibit?**

(A) Chief and mucous neck cells, but not parietal cells
(B) G, ECL, parietal, and chief cells
(C) Foveolar, mucous neck, and D cells, but not G cells
(D) S, I, and K cells of the duodenum and the jejunum
(E) Only parietal cells, acting through their H2 receptors

**19** A 52-year-old man with chronic pancreatitis has bulky, greasy stools and weight loss. Fecal elastase is very low.

**Which treatment best replaces the missing step?**

(A) Lactase replacement
(B) Intrinsic factor tablets
(C) Bile salt supplements
(D) A gluten-free diet
(E) Pancreatic enzyme replacement

**20** A 50-year-old woman has sudden severe epigastric pain radiating to her back with vomiting. Lipase is 9 times the upper limit of normal, ALT is 320 U/L, and ALP is raised. She drinks alcohol rarely. Ultrasound shows gallstones.

**Where is the obstructing stone most likely located?**

(A) Gallbladder fundus
(B) Hepatopancreatic ampulla (of Vater)
(C) Cystic duct
(D) Right hepatic duct at the porta hepatis
(E) Common hepatic duct near the hilum

**21** A 71-year-old woman has fever of 39.2°C, right upper quadrant pain, and jaundice. Over several hours she becomes confused, and her blood pressure falls to 82/48 mm Hg despite IV fluids. Ultrasound shows a dilated common bile duct with a stone.

**Which is the most important definitive step after starting IV ABX and resuscitation?**

(A) HIDA scan
(B) Oral ABX and outpatient follow-up
(C) Ursodeoxycholic acid
(D) Elective cholecystectomy in 6 weeks
(E) Urgent ERCP for biliary drainage

**22** A student traces the urea cycle to its final step.

**Which enzyme releases urea, and what does it regenerate?**

(A) Arginase, regenerating citrulline
(B) Argininosuccinate lyase, regenerating ornithine
(C) Ornithine transcarbamylase, regenerating arginine
(D) Arginase, regenerating ornithine
(E) Argininosuccinate synthetase, regenerating aspartate

**23** A 78-year-old man has sudden, painless passage of a large volume of bright red blood per rectum. He is otherwise well, with a soft, nontender abdomen.

**Which is the most likely cause?**

(A) Anal fissure
(B) Ischemic colitis
(C) Diverticulosis
(D) Ulcerative colitis
(E) Internal hemorrhoids

**24** A 71-year-old man who smokes has 3 months of progressive dysphagia to solids and liquids and has lost 11 kg. Barium swallow shows a "bird's beak" narrowing at the gastroesophageal junction.

**Which is the most appropriate next step?**

(A) Trial of a calcium channel blocker
(B) Upper endoscopy
(C) Botulinum toxin injection
(D) Pneumatic dilation
(E) Esophageal manometry

**25** A 33-year-old woman with diarrhea-predominant IBS and persistent abdominal pain is started on low-dose amitriptyline at night.

**Which adverse effects is she most likely to experience?**

(A) Diarrhea and hypersalivation
(B) Hyperkalemia and gynecomastia
(C) Dry mouth and constipation
(D) Bradycardia and miosis
(E) Weight loss and insomnia

**26** A 62-year-old obese man with 20 years of poorly controlled GERD undergoes endoscopy. Biopsies from salmon-colored mucosa above the GE junction show columnar epithelium with goblet cells.

**This patient is at greatest risk of developing which of the following?**

(A) Squamous cell carcinoma of the upper esophagus
(B) Gastric MALT lymphoma
(C) Zenker diverticulum
(D) Adenocarcinoma of the distal esophagus
(E) Esophageal leiomyoma

**27** A 26-year-old woman at 14 weeks of pregnancy has periumbilical pain that has moved to the RLQ over 18 hours, with anorexia and a temperature of 38.0 °C.

**Which imaging is most appropriate first?**

(A) Ultrasound
(B) Abdominal x-ray
(C) CT AP with contrast
(D) Upright CXR
(E) Colonoscopy

**28** A 67-year-old man has epigastric pain, early satiety, and weight loss. A hard, fixed node is felt above his left clavicle.

**Which cancer does this sign most suggest?**

(A) Cholangiocarcinoma
(B) Metastatic gastric cancer
(C) Colorectal adenocarcinoma
(D) Hepatocellular carcinoma
(E) Carcinoid tumor

**29** A 24-year-old man has had recurrent sinusitis and two episodes of giardiasis. Serum IgA is undetectable, with normal IgG and IgM. He is booked for elective surgery.

**Which precaution is most important if he needs a blood transfusion?**

(A) Give IV immunoglobulin first
(B) Give only irradiated blood
(C) Avoid transfusion of platelets only
(D) Use blood products without IgA
(E) Premedicate with an antihistamine and use any blood

**30** A 58-year-old man with alcohol-related cirrhosis and ascites has fever, diffuse abdominal tenderness, and new confusion. Diagnostic paracentesis shows an ascitic neutrophil count of 520/μL.

**Which is the most appropriate treatment?**

(A) Large-volume paracentesis with albumin, without ABX
(B) IV third-generation cephalosporin plus IV albumin
(C) Oral rifaximin alone
(D) Emergency laparotomy
(E) Increase the diuretic dose

**31** A newborn with flat facial features vomits after the first feed. An abdominal x-ray shows a double bubble with no gas beyond.

**What is the mechanism of this defect?**

(A) Failed lateral septation of the foregut
(B) Failed migration of neural crest cells
(C) Failed obliteration of the vitelline duct
(D) Hypertrophy of the pyloric muscle
(E) Failed recanalization of the duodenum

**32** A 74-year-old woman with A-fib who is not anticoagulated has sudden, severe periumbilical pain that began 3 hours ago. She is writhing in pain, but her abdomen is soft with only mild tenderness. Serum lactate is elevated.

**Which is the most appropriate diagnostic test?**

(A) Colonoscopy with mucosal biopsy
(B) Barium enema with air contrast
(C) Plain abdominal x-ray
(D) Abdominal ultrasound
(E) CT angiography of the abdomen

**33** A 32-year-old woman with ulcerative colitis starts a drug that blocks lymphocytes from entering the gut mucosa but has little effect on immunity elsewhere.

**Which drug is this?**

(A) Azathioprine
(B) Vedolizumab
(C) Infliximab
(D) Tofacitinib
(E) Prednisone

**34** A 36-year-old woman with bilateral hilar lymphadenopathy and erythema nodosum has ALP 300 IU/L and GGT 240 IU/L, with normal bilirubin and normal bile ducts on ultrasound.

**Which process most likely explains her liver tests?**

(A) Gilbert syndrome
(B) Ischemic hepatitis after a low-flow state
(C) Choledocholithiasis
(D) Acute hepatitis B
(E) Infiltration by sarcoidosis

**35** A 34-year-old woman taking combined oral contraceptives, who has had three pregnancies, is found to have cholesterol gallstones.

**How does estrogen promote cholesterol gallstone formation?**

(A) It increases cholesterol secretion into bile
(B) It increases gallbladder contraction
(C) It causes hemolysis
(D) It increases bile salt synthesis
(E) It deconjugates bilirubin

**36** A 29-year-old woman taking oral contraceptives is diagnosed with Budd-Chiari syndrome on Doppler ultrasound. She has mild ascites, a normal mental status, INR 1.3 and no evidence of cirrhosis.

**Which is the most appropriate initial treatment?**

(A) Immediate liver transplantation
(B) Observation with repeat ultrasound in 6 months
(C) Transjugular intrahepatic portosystemic shunt (TIPS) before anticoagulation
(D) Anticoagulation and stopping the oral contraceptive
(E) Large-volume paracentesis alone

**37** A 31-year-old man has chronic diarrhea, bloating, and weight loss. He eats a regular diet that includes bread and pasta.

**Which is the best initial test for the most likely diagnosis?**

(A) Hydrogen breath test
(B) Upper endoscopy with duodenal biopsy
(C) IgA anti-tTG antibodies with total serum IgA
(D) Fecal elastase
(E) Stool ova and parasites on three separate samples

**38** A 40-year-old woman who becomes seasick on every boat trip is going on a 3-day cruise next week. She wants something to prevent it.

**Which option is most appropriate, and how does it work?**

(A) Oral metoclopramide, blocking D2 receptors and speeding gastric emptying
(B) Oral loratadine, blocking peripheral H1 receptors in the inner ear
(C) Transdermal scopolamine, blocking muscarinic input to the vestibular nuclei
(D) Oral aprepitant, blocking substance P in the area postrema
(E) Oral ondansetron, blocking 5-HT3 receptors on vagal afferents

**39** A 66-year-old man with painless jaundice has a CT showing a 2.5-cm mass in the head of the pancreas, with dilation of both the common bile duct and the pancreatic duct. There is no vascular invasion or metastasis.

**Which is the most appropriate definitive treatment?**

(A) Observation with repeat CT in 3 months
(B) Cholecystectomy
(C) Distal pancreatectomy
(D) Pancreaticoduodenectomy (Whipple procedure)
(E) ERCP stent alone as definitive treatment

**40** A 38-year-old woman has a 3-cm liver lesion found by chance on ultrasound. Contrast MRI shows peripheral nodular enhancement that gradually fills in toward the center. She has no symptoms or liver disease.

**Which is the most appropriate management?**

(A) Percutaneous biopsy
(B) Surgical resection
(C) Transarterial chemoembolization
(D) Reassurance; no biopsy or treatment needed
(E) Stop any oral contraceptives and repeat imaging in 1 month

**41** A 24-year-old man with allergic rhinitis is diagnosed with eosinophilic esophagitis after an episode of food impaction. He asks how to address the underlying cause of his condition.

**Which management targets the trigger of this condition?**

(A) Pneumatic dilation of the LES
(B) Long-term oral fluconazole
(C) Laparoscopic Nissen fundoplication for reflux
(D) Stopping all oral medications for 2 weeks
(E) Elimination of food allergens from the diet

**42** Several children at a childcare center develop fever, abdominal cramps, and small-volume bloody, mucoid stools. The outbreak spreads person to person despite no shared food source. Stool culture grows a non-motile, lactose-nonfermenting gram-negative rod that does not produce H2S.

**Which organism is most likely responsible?**

(A) Enterotoxigenic E. coli
(B) Bacillus cereus
(C) Shigella
(D) Norovirus
(E) Salmonella

**43** A 19-year-old woman is brought in 6 hours after taking 30 tablets of 500 mg paracetamol. She has mild nausea but is otherwise well. Her paracetamol level is above the treatment line on the nomogram.

**The antidote she should receive works mainly by which mechanism?**

(A) Replenishing glutathione to neutralize NAPQI
(B) Enhancing glucuronidation of paracetamol
(C) Inhibiting CYP2E1
(D) Binding paracetamol in the gut
(E) Increasing renal excretion of paracetamol

**44** A 61-year-old woman with vitiligo and Hashimoto thyroiditis has fatigue and numbness in her feet. Hemoglobin is 9.1 g/dL with an MCV of 112 fL. Antibodies to parietal cells and intrinsic factor are positive.

**Which region of the stomach is most likely to show atrophy?**

(A) Pylorus
(B) Antrum
(C) Body and fundus
(D) Cardia only
(E) Duodenal bulb

**45** A 61-year-old man using an inhaled corticosteroid for COPD and recently treated with broad-spectrum ABX has painful swallowing. Endoscopy shows adherent white plaques along the esophagus.

**Which finding would most likely be seen on a biopsy of the plaques?**

(A) Owl's eye inclusions in endothelial cells
(B) More than 15 eosinophils per high-power field
(C) Pseudohyphae and budding yeast
(D) Goblet cells in columnar epithelium
(E) Multinucleated giant cells

**46** Between meals, a wave of regular, propulsive contractions sweeps through the small intestine about every 90–120 minutes.

**What is this pattern?**

(A) The gastrocolic reflex triggered by gastric stretch
(B) Haustral segmentation
(C) The migrating motor complex
(D) Receptive relaxation
(E) Mass movement

**47** A 34-year-old woman with type 1 diabetes has early satiety and vomiting of food eaten hours earlier. Endoscopy shows no obstruction, and 4-hour scintigraphy shows delayed emptying. Diet changes have not helped.

**Which drug is first line?**

(A) Metoclopramide
(B) Ondansetron alone
(C) Octreotide
(D) Hyoscine butylbromide
(E) Loperamide

**48** A liver biopsy from a woman with acute liver failure in the third trimester of pregnancy is reviewed.

**Which histologic finding is most characteristic of acute fatty liver of pregnancy?**

(A) Macrovesicular steatosis with Mallory-Denk bodies
(B) PAS-positive globules in periportal hepatocytes
(C) Noncaseating granulomas
(D) Periportal fibrin deposits and hemorrhage
(E) Microvesicular steatosis

**49** A 28-year-old woman at 20 weeks' gestation has 1 day of abdominal pain, now worst in the right side of the abdomen, with nausea and a temperature of 38.0°C. WBC is 15000/μL.

**Which is the most appropriate initial imaging?**

(A) No imaging; observe for 48 hours and repeat the white cell count
(B) Barium enema
(C) CT AP with contrast
(D) Graded-compression ultrasound, followed by MRI if inconclusive
(E) Abdominal x-ray

**50** A 31-year-old man is brought in after a motorcycle collision. He has right upper quadrant tenderness and abrasions over the right lower ribs. Blood pressure is 128/76 mm Hg and heart rate 88/min, and they remain stable after 1 L of crystalloid.

**Which is the most appropriate next step?**

(A) Upper endoscopy
(B) Diagnostic peritoneal lavage
(C) Discharge with outpatient follow-up
(D) Emergency laparotomy
(E) CT abdomen with IV contrast

TEST A

# Answer sheet

Name _____ Date _____ Score _____ / 50

| | | | | |
|---|---|---|---|---|
| 1 A B C D E | 11 A B C D E | 21 A B C D E | 31 A B C D E | 41 A B C D E |
| 2 A B C D E | 12 A B C D E | 22 A B C D E | 32 A B C D E | 42 A B C D E |
| 3 A B C D E | 13 A B C D E | 23 A B C D E | 33 A B C D E | 43 A B C D E |
| 4 A B C D E | 14 A B C D E | 24 A B C D E | 34 A B C D E | 44 A B C D E |
| 5 A B C D E | 15 A B C D E | 25 A B C D E | 35 A B C D E | 45 A B C D E |
| 6 A B C D E | 16 A B C D E | 26 A B C D E | 36 A B C D E | 46 A B C D E |
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Fill one bubble for each question. The answer key at the start of the Test A answers uses the same layout.

# Test B

50 questions | 60 minutes | Start the timer

**1** A woman with selective IgA deficiency asks which condition she is most likely to develop alongside it.

**Which association is most common?**

(A) Celiac disease
(B) Achalasia
(C) Hepatocellular carcinoma
(D) Pyloric stenosis
(E) Hirschsprung disease

**2** A 27-year-old man has bloating, gas, and watery diarrhea within hours of drinking milk. His symptoms stop when he avoids dairy.

**Which treatment most directly addresses the deficient step?**

(A) A gluten-free diet
(B) Lactase replacement
(C) IM B12 replacement
(D) Oral vancomycin
(E) Pancreatic enzyme replacement

**3** A 19-year-old man is tackled during a football match and develops right upper quadrant pain. He also complains of pain at the tip of his right shoulder.

**The shoulder pain is referred through which nerve?**

(A) Vagus nerve
(B) Intercostal nerve
(C) Phrenic nerve
(D) Greater splanchnic nerve
(E) Long thoracic nerve

**4** A 68-year-old woman needs long-term NSAIDs and is started on misoprostol to protect her stomach.

**Which mechanism explains this protection?**

(A) It is an irreversible pump inhibitor that stops HCl at its source
(B) It is an antacid that neutralizes HCl within the gastric lumen
(C) It is a competitive H2 blocker that decreases cAMP in parietal cells
(D) It is a PG analog that decreases HCl and increases mucus
(E) It is a paste that binds proteins at the base of the ulcer

**5** A 52-year-old man is admitted with acute pancreatitis. Heart rate is 112/min, BUN is rising, and urine output is low.

**Which is the most important initial treatment?**

(A) Total parenteral nutrition
(B) Goal-directed IV fluids such as Hartmann solution
(C) Prophylactic IV carbapenem to prevent infected necrosis
(D) Urgent contrast CT
(E) Emergency necrosectomy

**6** A 23-year-old woman with acute liver failure after a paracetamol overdose becomes sweaty and less responsive in the ICU. Her INR is 4.2.

**Which should be checked immediately?**

(A) Thyroid-stimulating hormone
(B) Serum lipase
(C) Ceruloplasmin
(D) Serum ferritin
(E) Blood glucose

**7** A 4-week-old first-born boy has had projectile, nonbilious vomiting after every feed for 3 days and is hungry afterward. A firm mass is felt in the right upper quadrant.

**Which metabolic picture is expected?**

- (A) Hyperchloremic metabolic acidosis
- (B) Hyperkalemic metabolic acidosis
- (C) Hypochloremic, hypokalemic metabolic alkalosis
- (D) High anion gap metabolic acidosis
- (E) Respiratory acidosis

**8** A 34-year-old man with ulcerative colitis has an elevated ALP of 480 U/L on routine bloods, with normal bilirubin. Ultrasound shows no gallstones. He is p-ANCA positive and AMA negative.

**Which test is most appropriate to confirm the diagnosis?**

- (A) Serum IgG4 alone
- (B) Liver biopsy
- (C) MRCP
- (D) HIDA scan
- (E) Repeat ultrasound in 6 months

**9** A 30-year-old man becomes nauseated whenever he reads in a moving car. His brother, who lost all labyrinthine function after an infection, never does.

**Which input explains why the brother is protected?**

- (A) Vagal afferents from the stomach
- (B) Baroreceptors in the carotid sinus
- (C) The cortex and thalamus
- (D) The vestibular system
- (E) The chemoreceptor trigger zone

**10** A 5-year-old boy had bloody diarrhea for 4 days after eating an undercooked hamburger. He is now pale with petechiae and decreased urine output. Bloods show hemoglobin 7.8 g/dL, platelets 42000/µL, creatinine 2.1 mg/dL and schistocytes on the blood smear.

**Which intervention should be avoided?**

- (A) Monitoring electrolytes and urine output
- (B) ABX therapy
- (C) Careful IV fluid management
- (D) Red blood cell transfusion if symptomatic
- (E) Dialysis if indicated

**11** A 58-year-old man has had untreated achalasia for 25 years, with a markedly dilated esophagus full of retained food.

**He is at increased risk of which malignancy?**

- (A) Leiomyosarcoma of the esophageal muscle wall
- (B) Squamous cell carcinoma of the esophagus
- (C) Hepatocellular carcinoma
- (D) Adenocarcinoma of the proximal stomach
- (E) Gastric MALT lymphoma

**12** A 69-year-old man in the ICU with cardiogenic shock is receiving high-dose noradrenaline. He develops abdominal distention and rising lactate. CT angiography shows diffuse narrowing of the mesenteric arteries without an embolus or thrombus.

**Which is the most likely diagnosis?**

- (A) Mesenteric venous thrombosis
- (B) Non-occlusive mesenteric ischemia
- (C) Chronic mesenteric ischemia
- (D) SMA embolus
- (E) Volvulus

**13** A 27-year-old woman has 8 months of recurrent lower abdominal cramping that improves after bowel movements, with alternating loose and hard stools. She has no weight loss, bleeding, or nocturnal symptoms. FBC, CRP, fecal calprotectin, and celiac serology are normal.

**Which is the most appropriate next step?**

- (A) Reassurance and a trial of a low-FODMAP diet
- (B) CT AP
- (C) Empiric metronidazole for small bowel overgrowth
- (D) Start oral prednisone for presumed IBD
- (E) Colonoscopy

**14** An 86-year-old woman with achalasia and severe heart failure is not a candidate for dilation or surgery. She receives an endoscopic injection into the lower esophageal sphincter.

**The injected agent works by which mechanism?**

- (A) Stimulating muscarinic receptors
- (B) Blocking L-type calcium channels in smooth muscle
- (C) Blocking acetylcholine release from nerve terminals
- (D) Inhibiting the proton pump
- (E) Releasing nitric oxide

**15** A 56-year-old man has 2 days of right upper quadrant pain, jaundice, dark urine, and itching. He is afebrile. Bilirubin is 5.8 mg/dL and ALP is 410 U/L, with mildly raised AST/ALT and normal lipase. Ultrasound shows gallstones and a common bile duct dilated to 10 mm.

**Which is the most appropriate next step?**

- (A) Laparoscopic cholecystectomy alone
- (B) ERCP with stone extraction
- (C) CT-guided liver biopsy
- (D) HIDA scan
- (E) Repeat LFTs in 2 weeks

**16** A 61-year-old man has had heartburn for years. Over the past 2 months he has had difficulty swallowing solid food and has lost 6 kg. His hemoglobin is 10.8 g/dL with a low MCV.

**Which is the most appropriate next step?**

- (A) 24-hour pH monitoring
- (B) H. pylori stool antigen test
- (C) Upper endoscopy with biopsy
- (D) 8-week trial of a PPI
- (E) Esophageal manometry

**17** A 79-year-old woman has small bowel obstruction. CT shows air in the biliary tree, dilated small bowel, and a 3.5-cm gallstone in the distal ileum. She is resuscitated.

**Which is the most appropriate surgical procedure?**

- (A) Emergency cholecystectomy and fistula repair as the first step
- (B) Colonoscopic stone removal
- (C) Enterolithotomy to remove the stone from the bowel
- (D) Nasogastric decompression alone
- (E) Right hemicolectomy

**18** A patient has a stone in the common bile duct and develops an infection of the obstructed bile.

**Which combination of findings makes up Charcot triad?**

- (A) Painless jaundice, palpable gallbladder, weight loss
- (B) Right upper quadrant pain, Murphy sign, fever
- (C) Jaundice, pruritus, pale stools
- (D) Fever, right upper quadrant pain, jaundice
- (E) Fever, hypotension, confusion

**19** A 34-year-old man with HIV who stopped antiretroviral therapy a year ago presents with 2 weeks of painful swallowing. His CD4 count is 62/μL. Examination shows white plaques on the buccal mucosa that scrape off, leaving an erythematous base. He is hemodynamically stable and can tolerate oral fluids.

**Which of the following is the most appropriate next step in management?**

- (A) Empiric oral fluconazole
- (B) Oral acyclovir
- (C) IV ganciclovir
- (D) High-dose oral omeprazole
- (E) Upper endoscopy with biopsy before any treatment

**20** A traveler takes loperamide for watery diarrhea and has no sedation.

**How does loperamide slow the gut?**

- (A) 5-HT4 agonism on enteric neurons, which increases ACh release
- (B) Inhibition of acetylcholinesterase
- (C) Somatostatin receptor agonism
- (D) μ-opioid agonism on enteric neurons, reducing ACh release
- (E) Blockade of M3 receptors on smooth muscle

**21** A 30-year-old man has intense sympathetic activation immediately after a road crash.

**Which effect on the stomach is expected, and through which fibers?**

- (A) Stimulation through efferent CN X fibers
- (B) Inhibition through efferent CN X fibers
- (C) Stimulation through sympathetic fibers
- (D) No effect, as only CN X fibers reach the stomach
- (E) Inhibition through thoracolumbar sympathetic fibers

**22** A 47-year-old man who drinks a bottle of vodka daily has fever, jaundice, and tender hepatomegaly after a week of heavy drinking. AST is 240 U/L, ALT is 95 U/L, GGT is elevated, and MCV is 104 fL.

**Which is the most likely diagnosis?**

- (A) Paracetamol toxicity
- (B) Ischemic hepatitis
- (C) Acute viral hepatitis A
- (D) MASLD
- (E) Alcoholic hepatitis

**23** A 35-year-old man is admitted with full-thickness burns over 45% of his body surface area. On hospital day 3, he has an episode of coffee-ground emesis.

**Which mechanism best explains this complication?**

- (A) Urease-mediated mucosal injury
- (B) Inhibition of prostaglandin synthesis
- (C) Increased vagal stimulation of parietal cells
- (D) Gastrin-secreting tumor
- (E) Hypovolemia leading to gastric mucosal ischemia

**24** A 63-year-old woman who takes naproxen daily develops sudden, severe epigastric pain that spreads across the abdomen. She is tachycardic, and the abdomen is rigid with rebound tenderness. An upright chest x-ray shows air under the right hemidiaphragm.

**In addition to IV fluids, IV PPI, and ABX, which is the most appropriate next step?**

- (A) Emergency surgical repair
- (B) Barium swallow to locate the perforation
- (C) Urgent upper endoscopy
- (D) Nasogastric tube and serial abdominal examinations only
- (E) H. pylori testing and observation

**25** A newborn has severe unconjugated hyperbilirubinemia and develops kernicterus. UGT activity is absent.

**Which is the most likely diagnosis?**

(A) Crigler-Najjar syndrome type I
(B) Rotor syndrome
(C) Crigler-Najjar syndrome type II
(D) Gilbert syndrome
(E) Dubin-Johnson syndrome

**26** A 28-year-old woman who has taken oral contraceptives for 10 years presents with sudden right upper quadrant pain and hypotension. CT shows a 7-cm liver mass with surrounding hemorrhage.

**Which is the most likely underlying lesion?**

(A) Angiosarcoma
(B) Focal nodular hyperplasia
(C) Hemangioma
(D) Hepatocellular carcinoma
(E) Hepatic adenoma

**27** A 74-year-old male smoker collapses with severe back and left flank pain. His blood pressure is 78/40 mm Hg, and a pulsatile mass is felt above the umbilicus.

**What is the most appropriate first investigation?**

(A) CT KUB for a ureteric stone
(B) CT angiogram of the chest, abdomen, and pelvis
(C) Upright chest x-ray
(D) Bedside ultrasound
(E) Serum lipase

**28** A 31-year-old woman at 36 weeks' gestation has nausea, abdominal pain, and jaundice. Blood pressure is 128/80 mm Hg. Two possible diagnoses are being considered: acute fatty liver of pregnancy and HELLP syndrome.

**Which finding most favors acute fatty liver of pregnancy?**

(A) Platelet count of 60000/µL with schistocytes
(B) Raised LDH
(C) Proteinuria
(D) Blood pressure over 160/110 mm Hg
(E) Hypoglycemia with prolonged INR

**29** A 52-year-old woman has a positive anti-HCV antibody on routine one-time screening. She feels well, and her liver enzymes are normal.

**Which is the most appropriate next step?**

(A) Start direct-acting antiviral therapy
(B) Repeat anti-HCV antibody in 6 months
(C) HCV RNA by PCR
(D) Reassure her that she is immune
(E) Liver biopsy

**30** A 72-year-old woman with long-standing pulmonary hypertension and severe tricuspid regurgitation has jugular venous distention, bilateral leg edema, and a tender, enlarged liver. Pressing on the liver increases her jugular venous pulsation. At autopsy, the cut surface of the liver shows dark red centrilobular areas alternating with pale tan areas.

**Which mechanism best explains the liver findings?**

(A) Raised central venous pressure transmitted to the hepatic veins
(B) Portal vein thrombosis
(C) Fatty infiltration of hepatocytes
(D) Thrombosis of the hepatic veins blocking outflow from the sinusoids
(E) Occlusion of the hepatic artery

**31** A 26-year-old man has his third episode in 6 months of severe nausea, vomiting, and abdominal cramps. He spends hours in hot showers, which help. He has used cannabis daily for 5 years. Between episodes he is well.

**Which of the following is the most effective long-term treatment?**

(A) Amitriptyline prophylaxis for migraine
(B) Pyloric botulinum injection
(C) Stopping cannabis
(D) A scopolamine patch before each episode
(E) Daily ondansetron

**32** A 22-year-old man has mild, fluctuating jaundice with conjugated hyperbilirubinemia and normal enzymes. A liver biopsy done for another reason shows a black liver.

**Which condition is most likely?**

(A) Rotor syndrome
(B) Dubin-Johnson syndrome
(C) Hemochromatosis
(D) Gilbert syndrome
(E) Crigler-Najjar syndrome

**33** A 78-year-old woman with a history of gallstones has 3 days of colicky abdominal pain, distention, vomiting, and no passage of stool or gas. CT shows air in the biliary tree, dilated small bowel loops, and a 3-cm calcified stone in the right lower quadrant.

**Where is the stone most likely lodged?**

(A) Duodenojejunal flexure
(B) Pylorus
(C) Sigmoid colon
(D) Ileocecal valve
(E) Ampulla of Vater

**34** A 42-year-old woman with obesity has had three episodes of epigastric pain radiating to her back, each starting about an hour after a large fried meal and settling within 3 hours. She is pain-free today, afebrile, with a soft, nontender abdomen. LFTs and white cell count are normal. Ultrasound shows several mobile, shadowing gallstones and a normal gallbladder wall.

**Which is the most appropriate management?**

(A) No treatment; reassure
(B) IV ABX and admission for emergency surgery
(C) Elective laparoscopic cholecystectomy
(D) HIDA scan
(E) ERCP with sphincterotomy

**35** A 58-year-old man with celiac disease who has not followed a gluten-free diet presents with weight loss, abdominal pain, and a small bowel perforation. Pathology of the resected segment shows a T-cell malignancy.

**Which is the most likely diagnosis?**

(A) Colorectal adenocarcinoma
(B) Carcinoid tumor
(C) Extranodal marginal zone (MALT) lymphoma
(D) Burkitt lymphoma
(E) Enteropathy-associated T-cell lymphoma

**36** A 57-year-old man with untreated hereditary hemochromatosis develops fever, hypotension, and hemorrhagic bullae on his legs 1 day after eating raw oysters.

**Which organism is most likely responsible?**

(A) Staphylococcus aureus
(B) Vibrio cholerae
(C) Bacillus cereus
(D) Vibrio vulnificus
(E) Listeria monocytogenes

**37** A 22-year-old man has 18 hours of abdominal pain that began around the umbilicus and moved to the right lower quadrant, with anorexia and a low-grade fever. Lying on his left side, he has pain when his right hip is passively extended.

**This finding suggests the inflamed appendix lies in which position?**

(A) Anterior, just under McBurney point
(B) Retrocecal, against the iliopsoas muscle
(C) Left lower quadrant
(D) Pelvic, against the obturator internus
(E) Subhepatic

**38** A 19-year-old traveler has had high fever for 3 weeks. He suddenly develops severe abdominal pain, and CT shows free air near the distal ileum.

**Which structure is most likely to have perforated?**

(A) A duodenal ulcer
(B) The appendix
(C) A bleeding, then perforated Meckel diverticulum
(D) A sigmoid diverticulum
(E) A necrotic Peyer patch

**39** A baby is born to a mother who is HBsAg positive. The baby is healthy, at term, and weighs 3.4 kg.

**Which is the most appropriate management for the newborn?**

- (A) HBIG only
- (B) Tenofovir for the newborn for 6 months
- (C) Hepatitis B vaccine at 2 months of age
- (D) Hepatitis B vaccine plus HBIG within 12 hours of birth
- (E) Test the newborn for HBsAg at birth and treat only if positive

**40** A 3-day-old boy has not passed meconium and has a distended abdomen. A contrast enema suggests a narrow rectosigmoid with dilated bowel above.

**Which test confirms the diagnosis?**

- (A) Abdominal ultrasound for a double bubble
- (B) Sweat chloride test
- (C) Stool calprotectin
- (D) 4-hour gastric emptying scintigraphy
- (E) Rectal suction biopsy

**41** A student traces cholecalciferol from the skin to its active form.

**What is the correct order?**

- (A) Kidney, then liver, then skin
- (B) Liver, then skin, then kidney
- (C) Skin, then kidney, then liver
- (D) Liver, then kidney, then skin
- (E) Skin, then liver, then kidney

**42** A 70-year-old woman with vitamin D deficiency has low serum calcium despite a normal calcium intake.

**Which best explains her low calcium?**

- (A) Decreased absorption in the duodenum
- (B) Decreased absorption in the ileum
- (C) Increased uptake through DMT1
- (D) Decreased binding to haptocorrin
- (E) Increased breakdown by trypsin

**43** A researcher follows labeled iron from the duodenal lumen into the blood.

**Which sequence of proteins does it use?**

- (A) Ferroportin, DMT1, then transferrin
- (B) DMT1, ferroportin, then transferrin
- (C) SGLT1, GLUT2, then transcobalamin
- (D) CTR1, ATP7A, then ceruloplasmin
- (E) Transferrin, DMT1, then ferroportin

**44** A physiology lecture explains why the liver tolerates occlusion of the hepatic artery better than many other organs tolerate loss of their arterial supply.

**Which vessel provides most of the liver's blood flow?**

- (A) Portal vein
- (B) Inferior vena cava
- (C) Splenic artery
- (D) Hepatic artery proper
- (E) Hepatic veins

**45** A 26-year-old woman has bloating, foul-smelling stools, iron-deficiency anemia, and an itchy vesicular rash on her elbows and knees. Duodenal biopsy shows villous atrophy, crypt hyperplasia, and intraepithelial lymphocytes.

**This condition is most strongly associated with which HLA types?**

- (A) HLA-DR5
- (B) HLA-B27
- (C) HLA-DQ2 and DQ8
- (D) HLA-DR3 and DR4
- (E) HLA-DR2

**46** A 26-year-old woman has fatigue and yellow eyes. Her bilirubin is 52 µmol/L with direct bilirubin 5 µmol/L, hemoglobin 92 g/L, and normal AST, ALT, and ALP.

**Which tests should be obtained next?**

- (A) MRCP
- (B) FBC and blood film
- (C) Hepatitis serology
- (D) AMA
- (E) No tests, as this is Gilbert syndrome

**47** An 18-month-old boy has painless rectal bleeding. A technetium scan lights up a focus in the right lower abdomen.

**Failure of which structure to obliterate caused this?**

- (A) The buccopharyngeal membrane
- (B) The ductus venosus
- (C) The urachus between the bladder and umbilicus
- (D) The vitelline (omphalomesenteric) duct
- (E) The cloacal membrane

**48** A 59-year-old woman being treated for diverticulitis has persistent fever and pain after 3 days of IV ABX. CT shows a 6-cm pericolic fluid collection with an enhancing rim and no free air.

**Which is the most appropriate next step?**

- (A) Colonoscopy
- (B) Continue IV ABX alone and repeat CT in 1 week
- (C) Increase dietary fiber
- (D) CT-guided percutaneous drainage plus ABX
- (E) Emergency colectomy with end colostomy

**49** A 30-year-old man has bloody diarrhea with tenesmus. Colonoscopy shows continuous inflammation from the rectum to the splenic flexure. His alkaline phosphatase is elevated, and MRCP shows multifocal strictures and dilations ("beading") of the intra- and extrahepatic bile ducts.

**Besides colorectal cancer, this patient is at greatest risk for which malignancy?**

- (A) Cholangiocarcinoma
- (B) Gastric MALT lymphoma
- (C) Small-bowel lymphoma
- (D) Hepatocellular carcinoma
- (E) Pancreatic adenocarcinoma

**50** A 45-year-old man from rural Southeast Asia, who often eats raw freshwater fish, presents with painless jaundice and weight loss. Imaging shows a mass at the confluence of the right and left hepatic ducts.

**Which organism most likely increased his risk for this tumor?**

- (A) Clonorchis sinensis
- (B) Helicobacter pylori
- (C) Entamoeba histolytica
- (D) Hepatitis A virus
- (E) Schistosoma haematobium

TEST B

# Answer sheet

Name _____ Date _____ Score _____ / 50

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|---|---|---|---|---|
| 1 A B C D E | 11 A B C D E | 21 A B C D E | 31 A B C D E | 41 A B C D E |
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Fill one bubble for each question. The answer key at the start of the Test B answers uses the same layout.

# Test C

50 questions | 60 minutes | Start the timer

**1** A 27-year-old woman with ulcerative colitis has a severe flare with 10 bloody stools per day. Her temperature is 39.0°C and heart rate 124/min, and her abdomen is distended and tender. Abdominal x-ray shows a transverse colon dilated to 8 cm.

**Which medication should be avoided?**

- (A) Loperamide
- (B) IV fluids
- (C) Broad-spectrum IV ABX
- (D) Subcutaneous heparin for DVT prophylaxis
- (E) IV methylprednisolone

**2** A 41-year-old man with known chronic hepatitis B, who injects drugs, has a sudden flare of hepatitis with jaundice. HBsAg is positive, IgM anti-HBc is negative, and IgG anti-HBc and anti-HDV are positive.

**Which statement about his new infection is most accurate?**

- (A) It is an HDV superinfection and is likely to become chronic
- (B) It is an HDV coinfection and will likely resolve
- (C) It can be prevented by the hepatitis C vaccine
- (D) It is reactivation of HBV alone
- (E) It is acute hepatitis A

**3** A physiologist notes that toxins in the blood, gastric irritation, motion, and fear can all cause vomiting.

**Where do these different inputs converge?**

- (A) The vestibular nuclei
- (B) The nucleus tractus solitarius
- (C) The dorsal motor nucleus of the vagus
- (D) The hypothalamus
- (E) The nucleus ambiguus

**4** A 68-year-old man who smokes has 6 weeks of progressive jaundice, itching, and a 7-kg weight loss, with no abdominal pain. On examination, he has a smooth, nontender mass below the right costal margin that moves with respiration. ALP and conjugated bilirubin are markedly raised.

**Which is the most likely cause?**

- (A) Pancreatic head or other periampullary malignancy
- (B) Choledocholithiasis
- (C) Mirizzi syndrome from a stone in the cystic duct
- (D) Hepatitis A
- (E) Acute cholecystitis with a stone in the gallbladder neck

**5** A newborn drools excessively and chokes with the first feed. A nasogastric tube cannot be passed into the stomach. The pregnancy was complicated by polyhydramnios.

**Which anatomy is most likely?**

- (A) Atresia without any fistula
- (B) A blind upper esophageal pouch with a distal tracheoesophageal fistula
- (C) An annular pancreas
- (D) A duodenal web
- (E) An isolated tracheoesophageal fistula with a fully intact esophagus

**6** A 50-year-old woman develops fever, eosinophilia, and severe hepatitis 1 week after her second general anesthetic in 3 months. Both used a halogenated inhaled anesthetic.

**Which mechanism best explains her liver injury?**

- (A) Copper accumulation
- (B) Hypoperfusion of the liver during repeated prolonged surgery
- (C) Bile duct injury during surgery
- (D) Immune reaction to drug-modified liver proteins
- (E) Excess NAPQI formation

**7** A histology slide of the distal ileum shows a dome of lymphoid tissue covered by epithelial cells with short, irregular microfolds instead of a brush border. Macrophages sit in pockets at their bases.

**What is the main function of these epithelial cells?**

(A) Producing mucus that coats the villi
(B) Transporting luminal antigens to immune cells by transcytosis
(C) Absorbing bile salts for return to the liver
(D) Releasing serotonin onto vagal afferents
(E) Secreting mucus and defensins into the crypt lumen to trap bacteria

**8** A 24-year-old woman with Wilson disease has finished initial copper chelation and needs maintenance treatment.

**Which drug works by blocking intestinal copper absorption?**

(A) Deferoxamine
(B) Cholestyramine
(C) D-penicillamine
(D) Trientine
(E) Zinc

**9** A 19-year-old man has a 6-month history of hand tremor, slurred speech, and new irritability. His AST and ALT are mildly elevated. Slit-lamp examination shows a golden-brown ring at the edge of each cornea. Serum ceruloplasmin is low.

**A defect in which process best explains his condition?**

(A) Intestinal iron absorption
(B) Secretion of alpha-1 antitrypsin from hepatocytes
(C) Conjugation of bilirubin
(D) Hepatic uptake of LDL cholesterol
(E) Hepatocyte copper excretion into bile

**10** A 55-year-old woman with primary biliary cholangitis on ursodeoxycholic acid has severe itching that keeps her awake. She is started on a drug that binds bile salts in the intestine and prevents their reabsorption.

**Which drug was started?**

(A) Orlistat
(B) Polyethylene glycol
(C) Ezetimibe
(D) Cholestyramine
(E) Octreotide

**11** A student traces bilirubin from heme to the blood.

**Which enzyme forms unconjugated bilirubin?**

(A) Biliverdin reductase
(B) UGT
(C) Glutathione reductase
(D) Heme oxygenase
(E) Alcohol dehydrogenase

**12** A 35-year-old woman eats a fatty meal. Her gallbladder contracts, pancreatic enzymes are released, and gastric emptying slows.

**Which cell and hormone best explain these effects?**

(A) L cells secreting PYY
(B) S cells secreting secretin
(C) I cells secreting CCK
(D) G cells secreting gastrin
(E) K cells secreting GIP

**13** A 58-year-old man had a cholecystectomy 6 years ago. He now has jaundice from a stone in the common bile duct. The stone is soft and brown, and bile culture grows E. coli.

**Which best explains how this stone formed?**

(A) Bacterial deconjugation of bilirubin within the bile duct
(B) Calcium oxalate precipitation
(C) Excess bile salt secretion
(D) Cholesterol supersaturation in the gallbladder
(E) Chronic hemolysis

**14** A 28-year-old man wants fast relief of occasional heartburn after large meals.

**Which drug class acts on HCl already in the lumen?**

- (A) H2R antagonists
- (B) Misoprostol
- (C) PPIs
- (D) Sucralfate
- (E) Antacids

**15** A 78-year-old man in the ICU after hip surgery develops marked colonic distension. CT shows a dilated cecum and colon with no mechanical obstruction. Bowel rest and decompression have not helped.

**Which drug is first line?**

- (A) Neostigmine
- (B) Loperamide
- (C) Octreotide
- (D) Morphine
- (E) Hyoscine butylbromide

**16** A 23-year-old woman who smokes has crampy right lower quadrant pain, non-bloody diarrhea, and weight loss. She also has a perianal fistula. Colonoscopy shows patchy ulcers in the terminal ileum and ascending colon separated by normal mucosa, and the rectum is spared.

**Which biopsy finding is most likely?**

- (A) PAS-positive foamy macrophages
- (B) Caseating granulomas
- (C) Crypt abscesses limited to the mucosa
- (D) Villous atrophy with intraepithelial lymphocytes
- (E) Noncaseating granulomas with transmural inflammation

**17** A 30-year-old pregnant woman with mild heartburn is started on famotidine.

**Famotidine reduces acid secretion by which mechanism?**

- (A) Activating prostaglandin E1 receptors
- (B) Blocking H2 receptors, lowering cAMP in parietal cells
- (C) Irreversibly blocking the proton pump in parietal cells
- (D) Blocking muscarinic M3 receptors
- (E) Binding bile salts

**18** A 32-year-old woman meets diagnostic criteria for IBS. Her clinician explains that her abdominal pain partly reflects increased central amplification of signals from gut stretch receptors.

**Which condition is she most likely to have as well?**

- (A) Ulcerative colitis
- (B) Fibromyalgia
- (C) Primary biliary cholangitis
- (D) Celiac disease
- (E) Hemochromatosis

**19** A 63-year-old man has painless jaundice. Imaging shows strictures of the bile ducts and a diffusely enlarged, "sausage-shaped" pancreas. Serum IgG4 is markedly elevated, and no mass is seen.

**Which is the most appropriate treatment?**

- (A) Pancreaticoduodenectomy
- (B) Liver transplant
- (C) Ursodeoxycholic acid alone
- (D) Corticosteroids
- (E) Broad-spectrum ABX

**20** A 68-year-old man has tender hepatomegaly, ascites, bilateral leg edema, and distended neck veins that rise further with pressure over the liver. Bilirubin and AST/ALT are mildly raised.

**Which is the most appropriate next test?**

- (A) Anti-smooth muscle antibody
- (B) Liver biopsy
- (C) Echocardiography
- (D) Doppler ultrasound of the hepatic veins
- (E) MRCP

**21** A 61-year-old man with HCV cirrhosis has a 3-cm liver lesion on surveillance ultrasound. Multiphase CT shows arterial-phase enhancement with washout in the portal venous phase. AFP is 420 ng/mL.

**Which is the most appropriate next step to confirm the diagnosis?**

(A) PET scan
(B) Percutaneous biopsy
(C) Diagnostic laparoscopy
(D) Repeat ultrasound in 3 months
(E) No further test is needed

**22** A 33-year-old woman has had 10 days of intractable hiccups, nausea, and vomiting. Endoscopy and abdominal imaging are normal. Six months ago she had severe optic neuritis that left poor vision in one eye.

**Which test is most likely to explain her symptoms?**

(A) β-hCG
(B) Gastric emptying study
(C) Serum AQP4 antibody
(D) Serum lipase
(E) Urine cannabinoid screen

**23** A 41-year-old woman with a positive H. pylori stool antigen test had anaphylaxis to penicillin as a child. Local clarithromycin resistance is low.

**Which triple therapy regimen is most appropriate?**

(A) PPI + amoxicillin + rifabutin
(B) PPI + clarithromycin + metronidazole
(C) PPI alone for 8 weeks
(D) PPI + amoxicillin + levofloxacin
(E) PPI + amoxicillin + clarithromycin

**24** A 9-year-old girl has 1 day of periumbilical pain that has moved to the right lower quadrant, with vomiting and a temperature of 38.1°C. She is tender at McBurney point. WBC is 14500/µL.

**Which is the most appropriate initial imaging test?**

(A) Barium enema
(B) CT AP with contrast
(C) Right lower quadrant ultrasound
(D) Colonoscopy
(E) Erect and supine abdominal x-ray

**25** A 74-year-old woman with long-standing gallstones has air in the biliary tree on CT. She has no history of biliary surgery or ERCP.

**Which type of biliary fistula is most common?**

(A) Cholecystoduodenal
(B) Cholecystocolic
(C) Cholecystogastric
(D) Cholecystocutaneous
(E) Choledochoduodenal

**26** A 59-year-old man with a duodenal ulcer takes high-dose omeprazole. Sucralfate is added, but it provides little extra benefit.

**Which best explains this?**

(A) It needs a high pH to bind the ulcer
(B) It needs acid to polymerize into a paste
(C) It is neutralized by bicarbonate
(D) It acts on H2 receptors, now blocked
(E) Omeprazole blocks its target

**27** Two patients have chronic H. pylori infection. One strain carries a gene encoding a protein that is injected into gastric epithelial cells by a type IV secretion system.

**Strains with this virulence factor carry a higher risk of which outcome?**

(A) Gastric adenocarcinoma
(B) Eosinophilic esophagitis
(C) Pancreatic insufficiency
(D) Celiac disease
(E) Achalasia

**28** A 59-year-old woman recovered from ascending cholangitis after ERCP with removal of a common bile duct stone. Ultrasound shows more gallstones in her gallbladder.

**Which is the most appropriate step to prevent recurrence?**

(A) Repeat ERCP every 6 months
(B) No further treatment
(C) Long-term oral ABX
(D) Cholecystectomy
(E) Ursodeoxycholic acid only

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**29** A 45-year-old man's brother was recently diagnosed with HFE hereditary hemochromatosis. He feels well and asks to be tested.

**Which is the best initial test?**

(A) Transferrin saturation and serum ferritin
(B) Echocardiography
(C) MRI of the liver
(D) Liver biopsy with Prussian blue stain
(E) Serum ceruloplasmin

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**30** A researcher studies how IgA made by plasma cells in the lamina propria reaches the intestinal lumen.

**Which step carries dimeric IgA across the epithelium?**

(A) Passive leak of dimeric IgA between enterocytes through loosened tight junctions
(B) Uptake by M cells from the lumen
(C) Binding to pIgR on the basolateral surface, then transcytosis
(D) Release from mast cells after IgE cross-linking
(E) Binding to the neonatal Fc receptor on the apical surface

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**31** An embolus blocks the SMA in an older woman.

**Which embryologic region of the gut does this artery supply?**

(A) Cloaca
(B) Hindgut
(C) Foregut
(D) Midgut
(E) Vitelline duct

---

**32** A 19-year-old man with asthma and atopic dermatitis presents to the emergency department with a piece of steak stuck in his throat. He reports years of intermittent dysphagia to solids. Endoscopy shows concentric esophageal rings and linear furrows. Biopsy shows more than 15 eosinophils per high-power field.

**Which of the following is the most likely diagnosis?**

(A) Candida esophagitis
(B) Schatzki ring
(C) Achalasia
(D) Eosinophilic esophagitis
(E) Gastroesophageal reflux disease

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**33** After an emergency delivery for acute fatty liver of pregnancy, the mother asks why the condition happened.

**A deficiency in the fetus of which enzyme is most strongly associated?**

(A) Glucose-6-phosphatase
(B) Hexosaminidase A
(C) Long-chain 3-hydroxyacyl-CoA dehydrogenase (LCHAD)
(D) Medium-chain acyl-CoA dehydrogenase (MCAD)
(E) Ornithine transcarbamylase

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**34** A student asks how sympathetic and parasympathetic input differ in their effects on the gut.

**Which statement is correct?**

(A) Sympathetic input is excitatory, through the vagus; parasympathetic input from T8–L2 is inhibitory
(B) Both are excitatory
(C) Sympathetic input reaches only the esophagus
(D) Parasympathetic input reaches only the rectum, through the pudendal nerves
(E) Sympathetic input is inhibitory, from T8–L2; parasympathetic input is mostly excitatory

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**35** A 68-year-old man is resuscitated after a cardiac arrest. The next day his ALT is 6400 IU/L and AST 7200 IU/L, with bilirubin 30 µmol/L. He took no medications before the arrest and has no history of liver disease.

**Which of the following is the most appropriate management of his liver injury?**

- (A) ERCP to clear the bile duct
- (B) High-dose prednisolone
- (C) Hepatitis serology before supportive care
- (D) Treat the low-flow state
- (E) IV N-acetylcysteine for presumed paracetamol toxicity

**36** A 14-year-old boy with sickle cell disease has recurrent right upper quadrant pain after meals. An abdominal x-ray shows several radiopaque stones in the right upper quadrant.

**Which best explains the formation of these stones?**

- (A) Calcium oxalate precipitation
- (B) Bacterial deconjugation of bilirubin in infected bile ducts
- (C) Increased unconjugated bilirubin from chronic hemolysis
- (D) Increased bile salt secretion
- (E) Cholesterol supersaturation of bile

**37** A 66-year-old man in the ICU after major burns, on TPN and vasopressors, develops fever and leukocytosis. Ultrasound shows a distended gallbladder with a thickened wall and fluid around it, but no gallstones. He is too unstable for surgery.

**Which is the most appropriate management?**

- (A) Oral ursodeoxycholic acid
- (B) ERCP with sphincterotomy
- (C) Observation only
- (D) Elective cholecystectomy in 6 weeks
- (E) IV ABX and percutaneous cholecystostomy

**38** A 66-year-old man is in the ICU on mechanical ventilation for 4 days after a severe head injury. His platelet count is 48000/µL.

**Which intervention best reduces his risk of stress-related gastric bleeding?**

- (A) No intervention is needed
- (B) H. pylori testing
- (C) Routine daily upper endoscopy
- (D) Prophylactic misoprostol
- (E) Prophylactic PPI or H2 blocker

**39** A 56-year-old woman has jaundice, dark urine, and pale stools.

**Which finding best explains her dark urine?**

- (A) Stercobilin reaching the urine
- (B) Urobilin falling in the urine
- (C) Conjugated bilirubin reaching the urine
- (D) Unconjugated bilirubin reaching the urine
- (E) Biliverdin reaching the urine

**40** A 26-year-old woman with Crohn disease develops severe epigastric pain radiating to the back 4 weeks after starting a new medication. Lipase is 8 times normal. She does not drink alcohol, ultrasound shows no gallstones, and triglycerides and calcium are normal.

**Which medication is the most likely cause?**

- (A) Budesonide
- (B) Loperamide
- (C) Vitamin D
- (D) Azathioprine
- (E) Infliximab

**41** A 29-year-old woman is diagnosed with celiac disease on serology and duodenal biopsy. She asks how her condition will be monitored once she starts treatment.

**Which is the best way to monitor her response to a gluten-free diet?**

- (A) Rising HLA-DQ2 expression
- (B) Repeat D-xylose test every month
- (C) Falling IgA anti-tTG levels
- (D) Serial fecal elastase
- (E) Annual colonoscopy

**42** A 24-year-old woman is stabbed in the right upper abdomen. Her blood pressure is 78/40 mm Hg and heart rate 136/min despite 2 units of blood. A FAST exam shows free fluid in Morison pouch.

**Which is the most appropriate next step?**

(A) Repeat FAST in 30 minutes
(B) Emergency laparotomy
(C) CT abdomen with IV contrast
(D) Angioembolization
(E) Serial abdominal examinations

**43** A 60-year-old man with alcohol-related cirrhosis has moderate ascites. He has stopped drinking. Creatinine and sodium are normal.

**Which is the most appropriate initial treatment?**

(A) Fluid restriction to 500 mL per day, with sodium intake left unrestricted
(B) Furosemide alone
(C) Immediate TIPS
(D) Weekly large-volume paracentesis
(E) Sodium restriction and spironolactone, with furosemide added as needed

**44** Two hours after eating leftover fried rice from a takeaway shop, a 25-year-old man develops nausea and repeated vomiting. His symptoms resolve within 24 hours.

**Which organism is most likely responsible?**

(A) Staphylococcus aureus
(B) Bacillus cereus
(C) Salmonella enteritidis
(D) Enterotoxigenic E. coli
(E) Clostridium perfringens

**45** A researcher follows labeled copper from the duodenal lumen across an enterocyte and into the blood.

**Which pair of transporters does it use?**

(A) DMT1, then ferroportin
(B) CTR1, then ATP7A
(C) SGLT1, then GLUT2
(D) PepT1, then ATP7B
(E) ATP7B, then CTR1

**46** A 38-year-old woman has dysphagia to solids and liquids and occasional regurgitation. Barium swallow suggests achalasia, and upper endoscopy shows no mass at the gastroesophageal junction.

**Which test is the gold standard for confirming the diagnosis?**

(A) Esophageal manometry
(B) Endoscopic ultrasound
(C) CT chest
(D) 24-hour esophageal pH monitoring
(E) Repeat barium swallow

**47** A 55-year-old man with chronic GERD has endoscopy showing Barrett esophagus without dysplasia. He is started on a daily PPI.

**Which is the most appropriate surveillance plan?**

(A) Esophagectomy
(B) No further endoscopy is needed
(C) Repeat endoscopy every 3 months
(D) Annual barium swallow
(E) Repeat endoscopy every 3–5 years

**48** A 22-year-old man is tackled during a rugby match and develops left upper quadrant pain. Twenty minutes later he also reports pain at the tip of his left shoulder, which he did not injure. His pulse is 118/min and blood pressure 96/60 mm Hg.

**Which mechanism best explains his shoulder pain?**

(A) Radicular pain from a lower cervical nerve root
(B) Direct injury to the acromioclavicular joint
(C) Somatoparietal pain from the inflamed lateral abdominal wall
(D) Referred pain from nerves sharing a spinal level
(E) Visceral pain from stretch of a hollow organ wall

**49** A 67-year-old woman with postprandial abdominal pain, fear of eating, and a 10-kg weight loss has severe atherosclerotic stenosis of the SMA and celiac trunk on CT angiography.

**Which is the most appropriate treatment?**

- (A) Long-term warfarin alone
- (B) Small frequent meals with nutritional support only
- (C) Endovascular angioplasty and stenting
- (D) Proton pump inhibitor
- (E) Cholecystectomy

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**50** A 72-year-old man with a lifelong low-fiber diet and chronic constipation is found to have many outpouchings in the sigmoid colon on colonoscopy.

**Which statement best describes these outpouchings?**

- (A) True diverticula containing all layers of the bowel wall, including the muscularis externa
- (B) Full-thickness herniation through the linea alba
- (C) Outpouchings of the posterior pharynx above the cricopharyngeus
- (D) False diverticula of mucosa and submucosa herniating at vasa recta entry points
- (E) Congenital remnants of the vitelline duct

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TEST C

# Answer sheet

Name _____ Date _____ Score _____ / 50

| | | | | |
|---|---|---|---|---|
| 1 A B C D E | 11 A B C D E | 21 A B C D E | 31 A B C D E | 41 A B C D E |
| 2 A B C D E | 12 A B C D E | 22 A B C D E | 32 A B C D E | 42 A B C D E |
| 3 A B C D E | 13 A B C D E | 23 A B C D E | 33 A B C D E | 43 A B C D E |
| 4 A B C D E | 14 A B C D E | 24 A B C D E | 34 A B C D E | 44 A B C D E |
| 5 A B C D E | 15 A B C D E | 25 A B C D E | 35 A B C D E | 45 A B C D E |
| 6 A B C D E | 16 A B C D E | 26 A B C D E | 36 A B C D E | 46 A B C D E |
| 7 A B C D E | 17 A B C D E | 27 A B C D E | 37 A B C D E | 47 A B C D E |
| 8 A B C D E | 18 A B C D E | 28 A B C D E | 38 A B C D E | 48 A B C D E |
| 9 A B C D E | 19 A B C D E | 29 A B C D E | 39 A B C D E | 49 A B C D E |
| 10 A B C D E | 20 A B C D E | 30 A B C D E | 40 A B C D E | 50 A B C D E |

Fill one bubble for each question. The answer key at the start of the Test C answers uses the same layout.

# Test A answers

Answer key | Explanations

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | D | 11 | D | 21 | E | 31 | E | 41 | E |
| 2 | B | 12 | A | 22 | D | 32 | E | 42 | C |
| 3 | B | 13 | E | 23 | C | 33 | B | 43 | A |
| 4 | D | 14 | E | 24 | B | 34 | E | 44 | C |
| 5 | A | 15 | A | 25 | C | 35 | A | 45 | C |
| 6 | A | 16 | B | 26 | D | 36 | D | 46 | C |
| 7 | C | 17 | A | 27 | A | 37 | C | 47 | A |
| 8 | A | 18 | B | 28 | B | 38 | C | 48 | E |
| 9 | B | 19 | E | 29 | D | 39 | D | 49 | D |
| 10 | C | 20 | B | 30 | B | 40 | D | 50 | E |

**1** **D** **Bile salts entering the colon directly**

Biliary fistulas | Step 1

A cholecystocolic fistula lets bile bypass the small intestine, where bile salts are normally reabsorbed. Bile salts reaching the colon stimulate water secretion, causing bile acid diarrhea; bile also fails to reach the small bowel, so fat absorption suffers.

**2** **B** **Anti-centromere antibody**

Autoimmune liver injury | Step 1

PBC is associated with other autoimmune diseases, including limited systemic sclerosis (CREST: calcinosis, Raynaud phenomenon, esophageal dysmotility, sclerodactyly, telangiectasia), which is linked to anti-centromere antibodies.

WHY NOT THE OTHERS

(D) Anti-Scl-70 is associated with diffuse systemic sclerosis.

**3** **B** **Increased estrogen from reduced hepatic clearance**

Acute liver failure and cirrhosis | Step 1

The cirrhotic liver clears estrogen poorly, and more androgen is converted to estrogen peripherally. The excess estrogen causes gynecomastia, testicular atrophy, palmar erythema, and spider angiomas.

**4** **D** **Its capillaries lack tight junctions, so it sits outside the blood-brain barrier**

Physiology | Emetic center | Step 1

The CTZ in the area postrema is a circumventricular organ whose capillaries lack tight junctions, so it can detect toxins in both blood and CSF. It lies on the floor of the fourth ventricle.

WHY NOT THE OTHERS

(B) The area postrema lies on the floor of the fourth ventricle.

(E) Vagal afferents carry signals from the gut to the NTS, not the other way.

**5** **A** **Unconjugated, as it is bound to albumin**

Physiology | Liver function | Step 2

Unconjugated bilirubin is water insoluble and bound to albumin, so it does not reach the urine.

**6** **A** **Pregnant patients**

PUD | Step 1

Misoprostol (a PGE1 analog) increases mucus and bicarbonate and reduces acid. It also causes uterine contractions and can induce abortion, so it

is contraindicated in pregnancy. Its other main side effect is diarrhea.

**7** **C** **Interferon-gamma release assay (or tuberculin skin test)**

IBD · Step 2

Infliximab is a TNF-α inhibitor. TNF-α is essential for maintaining granulomas, so blocking it can reactivate latent tuberculosis. Screen with an IGRA or tuberculin skin test (and for hepatitis B) before starting.

WHY NOT THE OTHERS

(B) TPMT activity is checked before azathioprine or 6-MP, not before anti-TNF therapy.

**8** **A** **IV ABX and laparoscopic cholecystectomy during this admission**

Cholecystitis · Step 2

This is acute calculous cholecystitis: prolonged RUQ pain, fever, Murphy sign, leukocytosis, typical ultrasound findings, and normal LFTs. Treat with supportive care and ABX, followed by early laparoscopic cholecystectomy, ideally within 72 hours.

WHY NOT THE OTHERS

(C) ERCP is for common bile duct stones; normal bilirubin and ALP make this unlikely.

(E) Cholecystostomy is reserved for patients too unwell for surgery or with acalculous cholecystitis.

**9** **B** **Sucrase, forming glucose and fructose**

Physiology · Digestion and absorption · Step 1

Sucrase converts sucrose to glucose and fructose, lactase converts lactose to glucose and galactose, and maltase converts maltose to glucose.

**10** **C** **Bismuth subcitrate**

Physiology · Gastric secretion · Step 2

Bismuth subcitrate coats the ulcer base, kills H. pylori, and increases PG and $HCO_3^-$.

**11** **D** **Hepatitis E virus**

Infectious liver injury · Step 1

HEV spreads by the fecal-oral route (contaminated water) and is usually self-limited, but in pregnancy, particularly the third trimester, it can cause fulminant hepatitis with high mortality. It is a naked +ssRNA hepevirus.

WHY NOT THE OTHERS

(C) HAV is also fecal-oral, but it is not associated with fulminant hepatitis in pregnancy.

**12** **A** **Oral rehydration, with loperamide if needed**

Diarrhea · Step 2

Mild traveler's diarrhea, most often from enterotoxigenic E. coli, is self-limited. Oral rehydration is the mainstay, with loperamide for symptom control if there is no fever or blood. ABX are reserved for moderate to severe cases.

**13** **E** **The 270° rotation of the midgut around the SMA**

Physiology · GI embryology · Step 1

The midgut rotates 270° counterclockwise around the SMA while herniated in weeks 6–10. Malrotation leaves bands and a mesentery that can twist into volvulus, and bilious vomiting, usually the first sign, needs urgent evaluation.

**14** **E** **Therapeutic phlebotomy**

Metabolic liver injury · Step 2

Regular phlebotomy is first-line: each unit of blood removes iron, and it is continued until ferritin is in the target range. Chelation is reserved for patients who cannot tolerate phlebotomy, such as those with anemia or heart failure.

WHY NOT THE OTHERS

(C) D-penicillamine chelates copper and is used for Wilson disease.

**15** **A** **Serum triglycerides of 2400 mg/dL**

Acute pancreatitis · Step 1

Hypertriglyceridemia (usually over 1000 mg/dL) is a cause of acute pancreatitis (the H in I GET SMASHED). Lipase breaks down the triglycerides into toxic free fatty acids that injure the pancreas. Milky (lipemic) serum is a clue. Treatment includes insulin infusion to lower triglycerides.

WHY NOT THE OTHERS

(D) Hypercalcemia (not low calcium) causes pancreatitis; low calcium is a complication of it.

**16** **B** **Intramuscular vitamin B12**

Gastritis · Step 2

Autoimmune gastritis destroys parietal cells, so intrinsic factor is lost and B12 cannot be absorbed normally. Parenteral (IM) B12 replacement

bypasses the problem; high-dose oral B12 is an alternative.

**WHY NOT THE OTHERS**

(A) Folic acid alone can correct the anemia while the neurologic damage from B12 deficiency progresses.

**17** **(A) It displaces toxic bile salts and increases bile flow**

Physiology · Liver function · Step 2

Ursodeoxycholic acid is a hydrophilic bile salt that displaces toxic hydrophobic bile salts and increases bile flow, slowing progression in PBC.

**18** **(B) G, ECL, parietal, and chief cells**

Physiology · Gastric secretion · Step 1

Somatostatin from D cells inhibits G, ECL, parietal, and chief cells. It acts through SSTR2, Gi, and decreased cAMP.

**19** **(E) Pancreatic enzyme replacement**

Physiology · Digestion and absorption · Step 2

Pancreatic insufficiency decreases pancreatic enzymes, leaving fat, protein, and carbohydrate undigested. Enzyme replacement treats it.

**20** **(B) Hepatopancreatic ampulla (of Vater)**

Choledocholithiasis · Step 1

Gallstone pancreatitis occurs when a stone blocks the ampulla, where the common bile duct and pancreatic duct meet, causing pancreatic enzyme activation. A cholestatic LFT pattern with high lipase and gallstones on ultrasound suggests a gallstone cause; it is the most common cause of acute pancreatitis.

**WHY NOT THE OTHERS**

(C) A cystic duct stone causes biliary colic or cholecystitis with normal LFTs and lipase.

**21** **(E) Urgent ERCP for biliary drainage**

Ascending cholangitis · Step 2

Charcot triad (fever, RUQ pain, jaundice) plus hypotension and confusion is Reynolds pentad: severe ascending cholangitis with septic shock. ABX alone are not enough; the infected, obstructed bile must be drained urgently, usually by ERCP.

**22** **(D) Arginase, regenerating ornithine**

Physiology · Liver function · Step 1

Arginase converts arginine to urea, releasing ornithine to restart the cycle.

**23** **(C) Diverticulosis**

Diverticular diseases · Step 1

Diverticulosis is the most common cause of major lower GI bleeding in older adults. It is typically painless, sudden, and large in volume because a vasa recta artery ruptures. Most bleeds stop on their own; colonoscopy finds and treats the source.

**WHY NOT THE OTHERS**

(A) An anal fissure causes painful, small-volume bleeding.

(B) Ischemic colitis causes pain with bleeding.

**24** **(B) Upper endoscopy**

Esophageal motility disorders · Step 2

Older age, a short history, and marked weight loss suggest pseudoachalasia from a tumor at the gastroesophageal junction. Endoscopy must exclude malignancy before achalasia is diagnosed or treated.

**25** **(C) Dry mouth and constipation**

IBS · Step 1

TCAs block muscarinic receptors, causing dry mouth, constipation, urinary retention, and blurred vision. They also cause sedation (via H1) and orthostasis (via α1). The constipating effect can be useful in IBS-D, and TCAs reduce visceral hypersensitivity.

**26** **(D) Adenocarcinoma of the distal esophagus**

GERD · Step 1

Barrett esophagus is intestinal metaplasia, in which nonciliated columnar epithelium with goblet cells replaces stratified squamous epithelium. It can progress through dysplasia to adenocarcinoma of the distal esophagus, so patients need endoscopic surveillance.

**WHY NOT THE OTHERS**

(A) Squamous cell carcinoma is linked to smoking and alcohol and typically arises in the upper or middle esophagus.

**27** **A** **Ultrasound**

Localizing abdominal pain | Step 2

For suspected appendicitis, CT AP with contrast is used in adults, but ultrasound comes first in children and pregnancy.

**WHY NOT THE OTHERS**

- (C) CT is avoided as the first test in pregnancy.
- (D) An upright CXR is for suspected perforation.
- (E) Colonoscopy has no role in suspected appendicitis.

**28** **B** **Metastatic gastric cancer**

Localizing abdominal pain | Step 1

A Virchow node, like a Sister Mary Joseph nodule or a Krukenberg tumor, points to metastatic gastric cancer.

**WHY NOT THE OTHERS**

- (E) Carcinoid syndrome causes flushing, wheezing, and tricuspid valve disease.

**29** **D** **Use blood products without IgA**

Physiology | Gut immunity | Step 2

Selective IgA deficiency, about 1 in 700, causes sinopulmonary infections and giardiasis. Anti-IgA antibodies can cause anaphylaxis after blood products containing IgA.

**WHY NOT THE OTHERS**

- (A) IV immunoglobulin contains IgA and could itself trigger a reaction.

**30** **B** **IV third-generation cephalosporin plus IV albumin**

Acute liver failure and cirrhosis | Step 2

An ascitic neutrophil count of 250/μL or more diagnoses spontaneous bacterial peritonitis. Treat with a third-generation cephalosporin (e.g. ceftriaxone or cefotaxime). IV albumin reduces the risk of hepatorenal syndrome. Long-term ABX prophylaxis follows.

**WHY NOT THE OTHERS**

- (D) SBP has no surgical source; laparotomy is for secondary peritonitis such as a perforation.

**31** **E** **Failed recanalization of the duodenum**

Physiology | GI embryology | Step 2

Duodenal atresia follows failed recanalization, usually below the ampulla, shows a double bubble, and occurs with Down syndrome in 30–40%.

**32** **E** **CT angiography of the abdomen**

Intestinal ischemia | Step 2

Pain out of proportion to examination in a patient with an embolic source suggests acute mesenteric ischemia. CT angiography is the diagnostic test of choice. Treatment is urgent revascularization (endovascular or surgical) with anticoagulation, plus resection of necrotic bowel.

**33** **B** **Vedolizumab**

Physiology | Gut immunity | Step 1

Vedolizumab binds the α4β7 integrin on gut-homing lymphocytes, blocking its interaction with MAdCAM-1, a gut-selective block on trafficking into the mucosa.

**WHY NOT THE OTHERS**

- (C) Infliximab blocks TNF-α throughout the body.

**34** **E** **Infiltration by sarcoidosis**

Reading liver function tests | Step 2

Infiltration, as in sarcoidosis, amyloidosis, or lymphoma, is a cause of the cholestatic pattern.

**WHY NOT THE OTHERS**

- (B) Ischemic hepatitis follows a low-flow state with ALT over 1000.
- (C) A duct stone usually dilates the ducts and causes pain.
- (D) Hepatitis B is ALT-predominant.

**35** **A** **It increases cholesterol secretion into bile**

Cholelithiasis | Step 1

Estrogen increases hepatic cholesterol uptake and synthesis (upregulating HMG-CoA reductase), so more cholesterol enters bile. Progesterone in pregnancy also slows gallbladder emptying. Both help explain the "female" and "fertile" Fs.

**36** **D** **Anticoagulation and stopping the oral contraceptive**

Vascular liver injury | Step 2

All patients with Budd-Chiari syndrome are anticoagulated, and the underlying cause is treated (here, stopping estrogen). Treatment then steps up if needed: angioplasty or stenting for short venous narrowing, then TIPS, then liver transplant for fulminant failure or cirrhosis.

**37** **C** **IgA anti-tTG antibodies with total serum IgA**

Malabsorption · Step 2

Screen for celiac disease with IgA anti-tTG while the patient is still eating gluten. Check total IgA because IgA deficiency (more common in celiac disease) gives false-negative results. Confirm with EGD and duodenal biopsy.

WHY NOT THE OTHERS

(B) Biopsy confirms the diagnosis but follows a positive serology.

**38** **C** **Transdermal scopolamine, blocking muscarinic input to the vestibular nuclei**

Physiology · Emetic center · Step 2

Scopolamine, given as a transdermal patch, blocks muscarinic receptors and inhibits input to the vestibular nuclei, preventing motion sickness. H1 antagonists that act in the brain also help.

WHY NOT THE OTHERS

(B) Second-generation antihistamines do not prevent motion sickness, likely because they act peripherally.

(E) 5-HT3 antagonists target chemotherapy, radiation, and postoperative vomiting, not vestibular input.

**39** **D** **Pancreaticoduodenectomy (Whipple procedure)**

Biliary malignancies · Step 2

A resectable tumor in the pancreatic head or periampullary region (the "double duct sign" shows both ducts blocked) is treated with pancreaticoduodenectomy (Whipple procedure), usually with chemotherapy. Stenting relieves jaundice but does not treat the cancer.

WHY NOT THE OTHERS

(C) Distal pancreatectomy is used for tumors of the pancreatic body and tail.

**40** **D** **Reassurance; no biopsy or treatment needed**

Liver tumors · Step 2

Peripheral nodular enhancement with centripetal fill-in is typical of a hemangioma, the most common benign liver tumor. Asymptomatic hemangiomas need no treatment, and biopsy is avoided because of bleeding risk.

**41** **E** **Elimination of food allergens from the diet**

Esophagitis · Step 2

Eosinophilic esophagitis is an allergic, food antigen-driven inflammation of the esophagus. Elimination diets remove the trigger; PPIs and swallowed topical corticosteroids are also first-line options.

WHY NOT THE OTHERS

(A) Pneumatic dilation of the LES treats achalasia.

**42** **C** **Shigella**

Diarrhea · Step 1

Shigella has a very low infectious dose (as few as 10 organisms), so it spreads easily person to person. It invades colonic mucosa, causing dysentery, and is non-motile and does not produce $H_2S$.

WHY NOT THE OTHERS

(E) Salmonella is motile, produces $H_2S$, and usually comes from poultry or eggs.

**43** **A** **Replenishing glutathione to neutralize NAPQI**

Iatrogenic and toxic liver injury · Step 1

In overdose, glucuronidation and sulfation are saturated, so more paracetamol is converted by CYP2E1 to the toxic metabolite NAPQI. Once glutathione is used up, NAPQI damages hepatocytes (zone 3 necrosis). N-acetylcysteine restores glutathione and is most effective within 8 hours.

WHY NOT THE OTHERS

(D) Binding the drug in the gut describes activated charcoal, which helps only within 1–2 hours.

**44** **C** **Body and fundus**

Gastritis · Step 1

Autoimmune gastritis targets parietal cells, which sit in the body and fundus. Their loss reduces HCl and intrinsic factor, leading to B12 deficiency (pernicious anemia, megaloblastic anemia, neuropathy). It causes fundal atrophy rather than erosions. H. pylori gastritis typically involves the antrum.

WHY NOT THE OTHERS

(B) The antrum is the classic site of H. pylori gastritis and contains G cells, not parietal cells.

**45** **C** **Pseudohyphae and budding yeast**

Esophagitis · Step 1

Candida esophagitis, the most common infectious esophagitis, shows pseudohyphae and budding yeast. Inhaled steroids, ABX, diabetes, and immunosuppression are risk factors.

**WHY NOT THE OTHERS**

- (A) Owl's eye inclusions suggest CMV.
- (D) Goblet cells in columnar epithelium describe Barrett esophagus.
- (E) Multinucleated giant cells suggest HSV.

**46** **C** **The migrating motor complex**

Physiology · Enteric nervous system · Step 1

The MMC, which depends on the ENS, has a quiet phase of about 70 minutes, an intermittent phase of 10–20 minutes, and 5 minutes of regular propulsion, recurring every 90–120 minutes.

**47** **A** **Metoclopramide**

Physiology · Enteric nervous system · Step 2

Gastroparesis is delayed gastric emptying without obstruction, often diabetic. Metoclopramide, which blocks D2 and stimulates 5-HT4, and erythromycin, a motilin agonist, are first line.

**WHY NOT THE OTHERS**

- (D) Hyoscine butylbromide reduces motility.
- (E) Loperamide slows transit.

**48** **E** **Microvesicular steatosis**

AFLP · Step 1

AFLP shows microvesicular steatosis: tiny fat droplets fill hepatocytes without displacing the nucleus, reflecting mitochondrial failure of fatty acid oxidation.

**WHY NOT THE OTHERS**

- (A) Macrovesicular steatosis with Mallory-Denk bodies suggests alcoholic liver disease.
- (D) Periportal fibrin deposits and hemorrhage suggest HELLP or pre-eclampsia.

**49** **D** **Graded-compression ultrasound, followed by MRI if inconclusive**

Appendicitis · Step 2

In pregnancy, ultrasound is used first to avoid radiation, then MRI without gadolinium if ultrasound is inconclusive. The appendix may be displaced upward by the uterus, so pain can be atypical. Delaying treatment risks perforation, which threatens the fetus.

**50** **E** **CT abdomen with IV contrast**

Traumatic liver injury · Step 2

A hemodynamically stable patient with suspected blunt abdominal trauma gets a contrast CT to identify and grade solid-organ injury. Most stable liver injuries are managed without surgery, with angioembolization if there is active bleeding.

**WHY NOT THE OTHERS**

- (D) Emergency laparotomy is for unstable patients with a positive FAST or peritonitis.

# Test B answers

Answer key | Explanations

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | A | 11 | B | 21 | E | 31 | C | 41 | E |
| 2 | B | 12 | B | 22 | E | 32 | B | 42 | A |
| 3 | C | 13 | A | 23 | E | 33 | D | 43 | B |
| 4 | D | 14 | C | 24 | A | 34 | C | 44 | A |
| 5 | B | 15 | B | 25 | A | 35 | E | 45 | C |
| 6 | E | 16 | C | 26 | E | 36 | D | 46 | B |
| 7 | C | 17 | C | 27 | D | 37 | B | 47 | D |
| 8 | C | 18 | D | 28 | E | 38 | E | 48 | D |
| 9 | D | 19 | A | 29 | C | 39 | D | 49 | A |
| 10 | B | 20 | D | 30 | A | 40 | E | 50 | A |

## 1 A Celiac disease

Physiology | Gut immunity | Step 1

Celiac disease is the condition most often found with IgA deficiency, and autoimmune disease occurs in 20–30%.

## 2 B Lactase replacement

Physiology | Digestion and absorption | Step 2

Lactase deficiency leaves lactose undigested, causing osmotic diarrhea and gas from bacterial digestion. Treatment is lactase replacement or a lactose-free diet.

## 3 C Phrenic nerve

Traumatic liver injury | Step 1

Blood or bile under the right hemidiaphragm irritates the diaphragmatic peritoneum, which is supplied by the phrenic nerve (C3–C5). Pain is referred to the C4 dermatome over the shoulder.

## 4 D It is a PG analog that decreases HCl and increases mucus

Physiology | Gastric secretion | Step 2

Misoprostol is a PG analog acting on EP receptors on parietal cells. It decreases HCl and increases mucus and HCO3−, giving gastroprotection in NSAID users.

## 5 B Goal-directed IV fluids such as Hartmann solution

Acute pancreatitis | Step 2

Third spacing causes hypovolemia in pancreatitis. Early goal-directed IV fluids (Hartmann solution is commonly used), pain control and early feeding are the core of treatment. ABX are only for proven infection, and enteral feeding is preferred over TPN.

### WHY NOT THE OTHERS

(C) Prophylactic ABX do not improve outcomes; they are reserved for infected necrosis.

## 6 E Blood glucose

Acute liver failure and cirrhosis | Step 1

Hypoglycemia is one of the "big 4" complications of acute liver failure, alongside encephalopathy, coagulopathy, and infection. The failing liver cannot make glucose or release it from glycogen. A sudden drop in consciousness needs an immediate glucose check.

## 7 C Hypochloremic, hypokalemic metabolic alkalosis

Physiology | GI embryology | Step 2

Pyloric stenosis causes projectile, nonbilious vomiting at 3–6 weeks with an olive in 60–80%, and the classic electrolyte picture is

hypochloremic, hypokalemic metabolic alkalosis. Treatment is pyloromyotomy.

**8** **C** **MRCP**

Autoimmune liver injury · Step 2

Primary sclerosing cholangitis affects intra- and extrahepatic bile ducts, so it is diagnosed by cholangiography: MRCP shows multifocal strictures and dilations ("beading"). Liver biopsy is reserved for suspected small-duct PSC with a normal MRCP.

WHY NOT THE OTHERS

(D) A HIDA scan evaluates acute cholecystitis.

**9** **D** **The vestibular system**

Physiology · Emetic center · Step 1

Motion sickness comes from mismatch between what the vestibular, visual, and proprioceptive systems report. People with total loss of labyrinthine function are immune. Lesioning the area postrema, by contrast, does not stop motion-induced vomiting.

WHY NOT THE OTHERS

(E) The CTZ is not needed for vomiting from motion.

**10** **B** **ABX therapy**

Diarrhea · Step 2

This is hemolytic uremic syndrome from Shiga toxin-producing E. coli (O157:H7). ABX, and antimotility agents such as loperamide, may increase toxin release and the risk of HUS, so they are avoided. Care is supportive: fluids, transfusion, and dialysis as needed.

**11** **B** **Squamous cell carcinoma of the esophagus**

Esophageal motility disorders · Step 1

Long-standing achalasia raises the risk of esophageal squamous cell carcinoma, probably from chronic irritation by stagnant food. Aspiration pneumonia is the other key complication.

**12** **B** **Non-occlusive mesenteric ischemia**

Intestinal ischemia · Step 1

Non-occlusive mesenteric ischemia follows low cardiac output and splanchnic vasoconstriction (shock, vasopressors, digoxin), without a blockage in the vessel. Treatment focuses on improving perfusion; intra-arterial vasodilators may help.

**13** **A** **Reassurance and a trial of a low-FODMAP diet**

IBS · Step 2

IBS is a disorder of gut-brain interaction diagnosed clinically (Rome V criteria) once alarm features are absent and basic tests are normal. First-line management is reassurance, education, and a low-FODMAP diet, with symptom-directed drugs as needed.

WHY NOT THE OTHERS

(E) Colonoscopy is not needed without alarm features in a patient under 45.

**14** **C** **Blocking acetylcholine release from nerve terminals**

Esophageal motility disorders · Step 1

Botulinum toxin cleaves SNARE proteins and blocks acetylcholine release from excitatory neurons, lowering LES tone. The effect wears off over months, so it is used in patients who are poor candidates for dilation or myotomy.

WHY NOT THE OTHERS

(B) Blocking L-type calcium channels is how calcium channel blockers such as nifedipine work.

**15** **B** **ERCP with stone extraction**

Choledocholithiasis · Step 2

Bilirubin over 4 mg/dL with a dilated CBD gives a high likelihood of choledocholithiasis, so proceed directly to ERCP to remove the stone, followed by cholecystectomy during the same admission. MRCP or endoscopic ultrasound is used when the likelihood is intermediate.

WHY NOT THE OTHERS

(A) Cholecystectomy alone leaves the CBD stone in place.

**16** **C** **Upper endoscopy with biopsy**

GERD · Step 2

Dysphagia, weight loss, and iron-deficiency anemia are alarm features that require endoscopy with biopsy to exclude malignancy, rather than an empiric PPI trial.

**17** **C** **Enterolithotomy to remove the stone from the bowel**

Biliary fistulas · Step 2

Gallstone ileus is treated by enterolithotomy (opening the bowel to remove the stone), which relieves the obstruction. Cholecystectomy and fistula repair are usually deferred, especially in frail patients, because a combined procedure carries higher risk.

**18** **D** **Fever, right upper quadrant pain, jaundice**

Ascending cholangitis · Step 1

Charcot triad is fever, right upper quadrant pain, and jaundice. Adding hypotension and altered mental status gives Reynolds pentad, which signals septic shock.

WHY NOT THE OTHERS

- (A) Painless jaundice with a palpable gallbladder suggests malignancy (Courvoisier law).
- (B) RUQ pain, Murphy sign, and fever without jaundice suggest acute cholecystitis.

**19** **A** **Empiric oral fluconazole**

Esophagitis · Step 2

Candida is the most common cause of infectious esophagitis. In an immunocompromised patient with odynophagia and oral thrush, empiric systemic fluconazole is appropriate. Endoscopy is reserved for patients who do not improve within about 72 hours.

WHY NOT THE OTHERS

- (B) Acyclovir treats HSV esophagitis (small "punched-out" ulcers); oral thrush points to Candida.
- (C) Ganciclovir treats CMV esophagitis (large, shallow linear ulcers).
- (E) EGD is indicated only if empiric antifungal therapy fails.

**20** **D** **μ-opioid agonism on enteric neurons, reducing ACh release**

Physiology · Enteric nervous system · Step 1

Loperamide acts on μ-opioid receptors on enteric neurons, reducing ACh release and propulsive peristalsis, with little reaching the brain.

**21** **E** **Inhibition through thoracolumbar sympathetic fibers**

Physiology · Gastric secretion · Step 1

Sympathetic fibers from the thoracolumbar outflow inhibit stomach activity, opposing the stimulation from efferent CN X.

**22** **E** **Alcoholic hepatitis**

Iatrogenic and toxic liver injury · Step 2

Alcoholic hepatitis typically shows an AST:ALT ratio above 2:1 with AST rarely above 300 U/L, raised GGT, and macrocytosis. ALT stays relatively low partly because it needs pyridoxal phosphate (vitamin B6), which is often deficient in heavy drinkers.

WHY NOT THE OTHERS

- (C) Viral hepatitis usually causes ALT-predominant rises, often well above 1000 U/L.
- (D) MASLD typically has ALT higher than AST.

**23** **E** **Hypovolemia leading to gastric mucosal ischemia**

Gastritis · Step 1

Curling ulcers follow severe burns. Massive fluid loss causes hypovolemia and gastric mucosal ischemia, which leads to erosions. Burn = Curling ("cooking"); brain = Cushing.

WHY NOT THE OTHERS

- (C) Increased vagal stimulation explains Cushing ulcers after brain injury or raised ICP.

**24** **A** **Emergency surgical repair**

PUD · Step 2

Free air with peritonitis means a perforated ulcer, which is a surgical emergency. Resuscitate with IV fluids, IV PPI, and broad-spectrum ABX, then proceed to surgery.

WHY NOT THE OTHERS

- (B) Barium leaking into the peritoneum causes severe peritonitis; water-soluble contrast is used if contrast is needed at all.
- (C) Endoscopy insufflates air and can worsen a known perforation.

**25** **A** **Crigler-Najjar syndrome type I**

Physiology · Liver function · Step 2

Crigler-Najjar syndrome type I has absent UGT activity, with severe neonatal jaundice and kernicterus. Type II is milder and responds to phenobarbital.

**26** **E** **Hepatic adenoma**

Liver tumors · Step 1

Hepatic adenomas are linked to oral contraceptives and anabolic steroids, and larger

ones (over 5 cm) can rupture and bleed. Management includes stopping estrogen and resecting large or bleeding adenomas.

**WHY NOT THE OTHERS**

(B) Focal nodular hyperplasia has a central stellate scar and rarely bleeds.

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## 27 (D) Bedside ultrasound

Localizing abdominal pain · Step 2

Pain with hypotension and a pulsatile mass suggests a ruptured AAA. Bedside ultrasound is used if the patient is unstable, and CT if stable.

**WHY NOT THE OTHERS**

(A) Shock with a pulsatile mass should not be attributed to a stone.

(B) CT is for a stable patient; he is too unstable to leave the resuscitation area.

(E) Lipase is for pancreatitis.

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## 28 (E) Hypoglycemia with prolonged INR

AFLP · Step 2

AFLP causes true liver failure, so hypoglycemia and coagulopathy (raised INR, low fibrinogen) are common. HELLP is defined by hemolysis, raised liver enzymes, and low platelets, usually with hypertension, and rarely causes hypoglycemia. Both are treated with delivery.

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## 29 (C) HCV RNA by PCR

Infectious liver injury · Step 2

A positive anti-HCV antibody shows exposure but not whether the infection is current; the antibody stays positive for life, even after cure. HCV RNA confirms active infection. If positive, assess fibrosis (e.g. APRI, FIB-4, or elastography) before treatment.

**WHY NOT THE OTHERS**

(D) Anti-HCV antibodies are not protective; reinfection is possible.

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## 30 (A) Raised central venous pressure transmitted to the hepatic veins

Vascular liver injury · Step 1

Right heart failure raises central venous pressure, which backs up into the hepatic veins and sinusoids. Zone 3 congestion and atrophy produce the mottled "nutmeg liver". Chronic congestion can lead to cardiac cirrhosis.

**WHY NOT THE OTHERS**

(B) Portal vein thrombosis causes portal hypertension without centrilobular congestion.

(D) Budd-Chiari syndrome also causes a nutmeg pattern, but JVD and hepatojugular reflux point to a cardiac cause.

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## 31 (C) Stopping cannabis

Physiology · Emetic center · Step 2

Years of cannabis use, cyclical vomiting relieved by hot showers, and wellness between episodes suggest cannabinoid hyperemesis syndrome. It resolves after stopping cannabis.

**WHY NOT THE OTHERS**

(A) Cyclic vomiting syndrome is linked with migraine, but the hot showers and daily cannabis point elsewhere.

(D) Scopolamine prevents motion sickness.

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## 32 (B) Dubin-Johnson syndrome

Reading liver function tests · Step 1

Dubin-Johnson syndrome, from MRP2 defects, raises conjugated bilirubin with a black liver; Rotor syndrome is similar without pigment.

**WHY NOT THE OTHERS**

(A) Rotor syndrome has no pigment.

(D) Gilbert syndrome raises unconjugated bilirubin.

(E) Crigler-Najjar syndrome raises unconjugated bilirubin.

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## 33 (D) Ileocecal valve

Cholecystitis · Step 1

Gallstone ileus: a large stone erodes through a cholecystoenteric fistula (usually cholecystoduodenal) and obstructs the narrowest part of the small bowel, the ileocecal valve. Rigler triad on imaging is pneumobilia, small bowel obstruction, and an ectopic gallstone.

**WHY NOT THE OTHERS**

(B) A stone obstructing the pylorus or duodenum causes Bouveret syndrome (gastric outlet obstruction), not distal small bowel obstruction.

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## 34 (C) Elective laparoscopic cholecystectomy

Cholelithiasis · Step 2

Recurrent biliary colic (transient cystic duct obstruction, normal LFTs, no fever) is an indication for elective laparoscopic cholecystectomy. Asymptomatic gallstones need no treatment, but symptomatic ones do, because attacks recur and complications such as cholecystitis can follow.

**WHY NOT THE OTHERS**

(A) Reassurance alone is for asymptomatic gallstones.

(E) ERCP removes common bile duct stones; her normal LFTs argue against choledocholithiasis.

## 35 (E) Enteropathy-associated T-cell lymphoma

Malabsorption · Step 1

Poorly controlled celiac disease raises the risk of enteropathy-associated T-cell lymphoma of the small bowel, as well as small bowel adenocarcinoma. A strict gluten-free diet lowers this risk.

## 36 (D) Vibrio vulnificus

Metabolic liver injury · Step 1

Iron overload favors iron-loving (siderophilic) bacteria such as Vibrio vulnificus, Yersinia, and Listeria. Vibrio vulnificus from raw shellfish can cause fulminant sepsis with hemorrhagic bullae, so patients are told to avoid raw shellfish.

## 37 (B) Retrocecal, against the iliopsoas muscle

Appendicitis · Step 1

The psoas sign (pain on passive hip extension) means a retrocecal appendix is irritating the iliopsoas. The obturator sign (pain on internal rotation of the flexed hip) suggests a pelvic appendix. Pain migrating from the umbilicus to the RLQ reflects visceral pain becoming parietal as inflammation reaches the peritoneum.

### WHY NOT THE OTHERS

(D) A pelvic appendix gives a positive obturator sign.

## 38 (E) A necrotic Peyer patch

Physiology · Gut immunity · Step 1

In typhoid, Salmonella targets M cells, and necrosis of Peyer patches can perforate the distal ileum, most often in the third week of fever.

## 39 (D) Hepatitis B vaccine plus HBIG within 12 hours of birth

Infectious liver injury · Step 2

Perinatal HBV transmission is highly likely to cause chronic infection (about 90%). Giving both the first vaccine dose and HBIG within 12 hours of birth prevents most infections.

## 40 (E) Rectal suction biopsy

Physiology · Enteric nervous system · Step 2

Hirschsprung disease follows failed migration of neural crest cells, leaving the rectosigmoid without ganglion cells in about 80%. It is diagnosed by rectal suction biopsy, with calretinin staining.

### WHY NOT THE OTHERS

(B) A sweat test investigates meconium ileus from cystic fibrosis.

## 41 (E) Skin, then liver, then kidney

Physiology · Liver function · Step 1

The skin makes cholecalciferol, the liver forms 25-OH cholecalciferol, and the kidney forms active calcitriol.

## 42 (A) Decreased absorption in the duodenum

Physiology · Digestion and absorption · Step 1

Calcium is absorbed in the duodenum, and vitamin D increases absorption.

## 43 (B) DMT1, ferroportin, then transferrin

Physiology · Digestion and absorption · Step 1

Iron enters the enterocyte through DMT1, reaches the blood through ferroportin, and is carried by transferrin.

## 44 (A) Portal vein

Vascular liver injury · Step 1

The liver has a dual blood supply: the portal vein provides about 75% of hepatic blood flow, and the hepatic artery about 25% (but roughly half the oxygen). This dual supply makes hepatic infarction uncommon.

## 45 (C) HLA-DQ2 and DQ8

Malabsorption · Step 1

Celiac disease is associated with HLA-DQ2 and DQ8. Gliadin (from wheat, rye, and barley) is deamidated by tissue transglutaminase and presented on these HLA molecules, driving a T-cell response that damages the mucosa. The rash is dermatitis herpetiformis. Iron deficiency reflects duodenal involvement.

### WHY NOT THE OTHERS

(B) HLA-B27 is linked to the seronegative spondyloarthropathies.

(D) HLA-DR3/DR4 is classically linked to type 1 diabetes.

**46** **B** **FBC and blood film**

Reading liver function tests · Step 1

Unconjugated bilirubin with anemia suggests hemolysis, investigated with FBC and blood film. Hemolysis rarely raises bilirubin above 4 mg/dL.

**WHY NOT THE OTHERS**

- (A) MRCP looks for duct disease, which raises conjugated bilirubin.
- (E) Gilbert syndrome has a normal FBC.

**47** **D** **The vitelline (omphalomesenteric) duct**

Physiology · GI embryology · Step 1

Meckel diverticulum follows incomplete obliteration of the vitelline duct, which closes in week 7; ectopic gastric mucosa causes painless bleeding, and the rule of 2s applies.

**48** **D** **CT-guided percutaneous drainage plus ABX**

Diverticular diseases · Step 2

Diverticular abscesses larger than about 3–4 cm are drained percutaneously alongside ABX. Surgery is reserved for failed drainage, perforation with peritonitis, or recurrent complicated disease.

**49** **A** **Cholangiocarcinoma**

IBD · Step 1

Primary sclerosing cholangitis is strongly associated with ulcerative colitis (p-ANCA positive) and greatly raises the risk of cholangiocarcinoma. It also raises colorectal cancer risk further, so patients with both need closer colonoscopic surveillance.

**50** **A** **Clonorchis sinensis**

Biliary malignancies · Step 1

A perihilar mass at the hepatic duct confluence (Klatskin tumor) is a cholangiocarcinoma. Liver flukes such as Clonorchis sinensis, acquired from raw freshwater fish, cause chronic bile duct inflammation and raise the risk. PSC and choledochal cysts are other risk factors.

**WHY NOT THE OTHERS**

- (E) Schistosoma haematobium is linked to squamous cell carcinoma of the bladder.

# Test C answers

Answer key | Explanations

| | | | | |
|---|---|---|---|---|
| 1 A | 11 A | 21 E | 31 D | 41 C |
| 2 A | 12 C | 22 C | 32 D | 42 B |
| 3 B | 13 A | 23 B | 33 C | 43 E |
| 4 A | 14 E | 24 C | 34 E | 44 B |
| 5 B | 15 A | 25 A | 35 D | 45 B |
| 6 D | 16 E | 26 B | 36 C | 46 A |
| 7 B | 17 B | 27 A | 37 E | 47 E |
| 8 E | 18 B | 28 D | 38 E | 48 D |
| 9 E | 19 D | 29 A | 39 C | 49 C |
| 10 D | 20 C | 30 C | 40 D | 50 D |

## 1 A Loperamide

IBD | Step 2

This is toxic megacolon. Antimotility agents (loperamide) and opioids worsen colonic dilation and can precipitate perforation. Management is bowel rest, IV fluids, IV steroids, ABX, and early surgical consultation. VTE prophylaxis is still needed because IBD is hypercoagulable.

## 2 A It is an HDV superinfection and is likely to become chronic

Infectious liver injury | Step 1

HDV needs HBsAg to form its envelope, so it only infects people with HBV. A negative IgM anti-HBc shows the HBV is chronic, so this is a superinfection, which is usually more severe and becomes chronic. In coinfection, both are acquired together and IgM anti-HBc is positive.

### WHY NOT THE OTHERS

(B) Coinfection would show a positive IgM anti-HBc, because the HBV infection would be acute.

(C) There is no hepatitis C vaccine; HDV is prevented by the hepatitis B vaccine.

## 3 B The nucleus tractus solitarius

Physiology | Emetic center | Step 1

The CTZ, vagal afferents, the vestibular system, and higher centers converge on the NTS, which begins the final common pathway and drives the central pattern generator in the medullary reticular formation.

## 4 A Pancreatic head or other periampullary malignancy

Biliary malignancies | Step 2

Courvoisier law: painless jaundice with a palpable, nontender gallbladder is unlikely to be caused by gallstones. A slowly growing tumor below the cystic duct lets a healthy gallbladder distend. Gallstones usually scar the gallbladder, so it cannot enlarge, and they cause pain.

### WHY NOT THE OTHERS

(B) CBD stones usually cause painful jaundice, and the gallbladder is typically shrunken from chronic inflammation.

## 5 B A blind upper esophageal pouch with a distal tracheoesophageal fistula

Physiology | GI embryology | Step 2

Esophageal atresia presents with drooling, choking, and failure to pass a nasogastric tube; type C, a proximal pouch with a distal fistula, accounts for 84%.

**6** **D** **Immune reaction to drug-modified liver proteins**

Iatrogenic and toxic liver injury · Step 1

Halothane hepatitis is an immune-mediated reaction: trifluoroacetylated liver proteins act as neoantigens, so severe injury usually follows repeat exposure. Fever and eosinophilia are clues to an immune drug reaction.

**7** **B** **Transporting luminal antigens to immune cells by transcytosis**

Physiology · Gut immunity · Step 1

M cells overlie Peyer patches, lack normal microvilli, and carry antigens across the epithelium by transcytosis to dendritic cells and macrophages waiting in a basolateral pocket.

WHY NOT THE OTHERS

(A) Mucus-secreting cells have a different structure; M cells sit in an epithelium with thin mucus.

**8** **E** **Zinc**

Physiology · Digestion and absorption · Step 2

Zinc blocks intestinal copper absorption and is used for maintenance. D-penicillamine and trientine are copper chelators.

**9** **E** **Hepatocyte copper excretion into bile**

Metabolic liver injury · Step 1

Wilson disease is caused by ATP7B mutations (autosomal recessive). ATP7B moves copper into bile and onto ceruloplasmin. Without it, copper builds up in the liver and then deposits in the basal ganglia (movement and psychiatric symptoms) and cornea (Kayser-Fleischer rings).

**10** **D** **Cholestyramine**

Autoimmune liver injury · Step 1

Cholestyramine is a bile acid sequestrant. It interrupts enterohepatic circulation, lowering bile salts that build up in cholestasis and cause pruritus. Give it apart from other drugs (including UDCA), because it binds them too.

WHY NOT THE OTHERS

(C) Ezetimibe blocks cholesterol absorption (NPC1L1) and is not used for pruritus.

**11** **A** **Biliverdin reductase**

Physiology · Liver function · Step 1

Biliverdin reductase forms unconjugated bilirubin, which albumin carries to the liver.

**12** **C** **I cells secreting CCK**

Physiology · Gastric secretion · Step 1

Fat and amino acids release CCK from I cells, which decreases HCl, slows gastric emptying, and causes gallbladder contraction and pancreatic enzyme release.

**13** **A** **Bacterial deconjugation of bilirubin within the bile duct**

Choledocholithiasis · Step 1

Brown pigment stones form within the bile ducts (primary CBD stones), where infection with bacteria producing β-glucuronidase deconjugates bilirubin, which precipitates with calcium. They can form years after cholecystectomy.

**14** **E** **Antacids**

Physiology · Gastric secretion · Step 2

Antacids neutralize HCl in the lumen, increasing gastric pH and giving fast, short-acting relief.

**15** **A** **Neostigmine**

Physiology · Enteric nervous system · Step 2

Acute colonic pseudo-obstruction (Ogilvie syndrome) affects sick inpatients. After excluding obstruction, neostigmine, an acetylcholinesterase inhibitor, is the first-line drug for refractory cases.

**16** **E** **Noncaseating granulomas with transmural inflammation**

IBD · Step 1

Crohn disease shows skip lesions, transmural inflammation (causing fistulas and strictures), noncaseating granulomas, and Th1-mediated inflammation. It most often involves the terminal ileum and spares the rectum. Smoking worsens Crohn disease.

WHY NOT THE OTHERS

(B) Caseating granulomas suggest intestinal tuberculosis.

(C) Inflammation limited to the mucosa and submucosa, with crypt abscesses, is typical of ulcerative colitis.

**17** **B** **Blocking H2 receptors, lowering cAMP in parietal cells**

GERD · Step 1

H2-receptor antagonists competitively block histamine H2 receptors on parietal cells, reducing cAMP and acid secretion. They are less potent than PPIs and suit milder symptoms.

**18** **B** **Fibromyalgia**

IBS · Step 1

IBS is a disorder of gut-brain interaction driven by visceral hypersensitivity, altered microbiome, and abnormal motility. It commonly coexists with fibromyalgia, anxiety, and depression.

**19** **D** **Corticosteroids**

Autoimmune liver injury · Step 2

IgG4-related sclerosing cholangitis often occurs with autoimmune pancreatitis and can mimic PSC or cancer. Unlike PSC, it responds well to corticosteroids.

**20** **C** **Echocardiography**

Vascular liver injury · Step 2

Hepatomegaly with jugular venous distention and a positive hepatojugular reflux points to congestive hepatopathy from right heart failure, so echocardiography is the key test. Budd-Chiari syndrome causes hepatomegaly and ascites without JVD.

**21** **E** **No further test is needed**

Liver tumors · Step 2

In a patient with cirrhosis, a lesion with arterial-phase enhancement and washout on multiphase CT or MRI is diagnostic of hepatocellular carcinoma; biopsy is not required. Biopsy is reserved for lesions with atypical imaging.

**22** **C** **Serum AQP4 antibody**

Physiology · Emetic center · Step 2

Area postrema syndrome, intractable nausea, vomiting, and hiccups, occurs when NMOSD attacks the AQP4-rich area postrema. With previous severe optic neuritis, AQP4 antibody testing is the key test.

WHY NOT THE OTHERS

(B) Gastroparesis does not explain hiccups or the history of optic neuritis.

(E) Cannabinoid hyperemesis causes cyclical vomiting relieved by hot showers.

**23** **B** **PPI + clarithromycin + metronidazole**

PUD · Step 2

In penicillin allergy, metronidazole replaces amoxicillin in triple therapy, giving PPI + clarithromycin + metronidazole.

WHY NOT THE OTHERS

(E) Amoxicillin is a penicillin and is contraindicated after anaphylaxis.

**24** **C** **Right lower quadrant ultrasound**

Appendicitis · Step 2

In children and pregnant patients, ultrasound is the first imaging test for suspected appendicitis to avoid radiation. CT with contrast is the most accurate test in adults, or if ultrasound is inconclusive.

**25** **A** **Cholecystoduodenal**

Biliary fistulas · Step 1

Air in the biliary tree without prior instrumentation suggests a biliary-enteric fistula. Cholecystoduodenal fistulas are the most common, because the gallbladder lies right next to the duodenum, and they are the usual route for gallstone ileus.

**26** **B** **It needs acid to polymerize into a paste**

Physiology · Gastric secretion · Step 2

Sucralfate polymerizes at pH under 4 into a paste that binds the ulcer base, so it works poorly with acid-suppressing drugs.

**27** **A** **Gastric adenocarcinoma**

PUD · Step 1

CagA-positive H. pylori strains cause more severe inflammation and carry a higher risk of peptic ulcers and gastric adenocarcinoma. VacA is another cytotoxin associated with more severe disease.

**28** **D** **Cholecystectomy**

Ascending cholangitis · Step 2

The gallbladder is the source of most CBD stones, so cholecystectomy (once she has recovered,

usually during the same admission) prevents recurrent choledocholithiasis and cholangitis.

**29** **A** **Transferrin saturation and serum ferritin**

Metabolic liver injury · Step 2

Transferrin saturation (above 45% is abnormal) with ferritin is the initial screen for iron overload. HFE genetic testing follows, which is also offered directly to first-degree relatives. Liver MRI or biopsy is reserved for unclear cases.

**30** **C** **Binding to pIgR on the basolateral surface, then transcytosis**

Physiology · Gut immunity · Step 1

Plasma cells secrete monomeric IgA, a J chain joins two monomers, and the dimer binds pIgR at the basolateral surface, crosses the enterocyte, and is released when pIgR is cleaved, leaving the secretory component on SIgA.

**31** **D** **Midgut**

Physiology · GI embryology · Step 1

The celiac artery supplies the foregut, the SMA the midgut, from the mid-duodenum to the proximal two-thirds of the transverse colon, and the IMA the hindgut.

**32** **D** **Eosinophilic esophagitis**

Esophagitis · Step 1

Eosinophilic esophagitis occurs in atopic patients (asthma, allergic rhinitis, atopic dermatitis) and is driven by food allergens. Classic findings are food impaction, "trachealized" rings and furrows, and at least 15 eosinophils/hpf. Management includes eliminating food allergens, PPIs, and swallowed topical steroids.

WHY NOT THE OTHERS

- (B) A Schatzki ring is a single distal mucosal ring and does not explain dense eosinophilic infiltration.
- (E) GERD can cause some eosinophils, but rings, furrows, atopy, and food impaction point to EoE.

**33** **C** **Long-chain 3-hydroxyacyl-CoA dehydrogenase (LCHAD)**

AFLP · Step 1

AFLP is linked to fetal LCHAD deficiency, a defect in mitochondrial beta-oxidation of long-chain fatty acids. Fatty acid metabolites from the fetus overwhelm the heterozygous mother's liver, causing microvesicular steatosis.

**34** **E** **Sympathetic input is inhibitory, from T8–L2; parasympathetic input is mostly excitatory**

Physiology · Enteric nervous system · Step 1

Sympathetic fibers from T8–L2 inhibit through prevertebral ganglia, while parasympathetic fibers in the vagus and pelvic nerves are mostly excitatory.

WHY NOT THE OTHERS

- (D) The pudendal nerves are somatic, supplying the external sphincter and puborectalis.

**35** **D** **Treat the low-flow state**

Reading liver function tests · Step 2

ALT over 1000 after shock, cardiac arrest, or heart failure suggests ischemic hepatitis, one of the causes of massive rises over 10000 IU/L. Treatment is restoring perfusion.

WHY NOT THE OTHERS

- (A) The pattern is hepatocellular, not cholestatic.
- (B) Prednisolone treats autoimmune or severe alcoholic hepatitis.
- (E) Without an ingestion, the timing after arrest points to ischemia.

**36** **C** **Increased unconjugated bilirubin from chronic hemolysis**

Cholelithiasis · Step 1

Chronic hemolysis (sickle cell disease, hereditary spherocytosis) increases bilirubin production. The extra bilirubin precipitates with calcium in the gallbladder as black pigment stones, which are often radiopaque.

WHY NOT THE OTHERS

- (B) Bacterial deconjugation produces brown pigment stones, which form in the bile ducts.

**37** **E** **IV ABX and percutaneous cholecystostomy**

Cholecystitis · Step 2

Acalculous cholecystitis occurs in critically ill patients from gallbladder stasis and ischemia, and carries a high risk of gangrene and perforation. Unstable patients are treated with ABX and percutaneous drainage (cholecystostomy).

**38** **E** **Prophylactic PPI or H2 blocker**

Gastritis · Step 2

Critically ill patients, especially those on mechanical ventilation, with coagulopathy, burns, or head injury (Cushing ulcers), are at risk of stress ulcer bleeding. Acid suppression with a PPI or H2 blocker is used as prophylaxis.

**39** **C** **Conjugated bilirubin reaching the urine**

Physiology · Liver function · Step 2

Only conjugated bilirubin reaches the urine, so dark urine points to conjugated hyperbilirubinemia. Stool is pale if bile is obstructed.

**40** **D** **Azathioprine**

Acute pancreatitis · Step 1

Drugs are the "D" in I GET SMASHED. Azathioprine is a classic cause of acute pancreatitis, usually within the first weeks of therapy. Others include valproate, thiazides, furosemide, sulfonamides, and didanosine.

**41** **C** **Falling IgA anti-tTG levels**

Malabsorption · Step 2

Treatment is a lifelong gluten-free diet (avoiding wheat, rye, and barley). Celiac antibodies such as IgA anti-tTG fall with good adherence, and persistently raised levels suggest ongoing gluten exposure.

WHY NOT THE OTHERS

(A) HLA type does not change; HLA-DQ2/DQ8 testing helps rule celiac disease out, not monitor it.

**42** **B** **Emergency laparotomy**

Traumatic liver injury · Step 2

A hemodynamically unstable trauma patient with a positive FAST needs emergency laparotomy to control bleeding. Unstable patients should not be sent to CT.

**43** **E** **Sodium restriction and spironolactone, with furosemide added as needed**

Acute liver failure and cirrhosis · Step 2

First-line ascites treatment is dietary sodium restriction plus spironolactone (blocking secondary hyperaldosteronism), often with furosemide. Large-volume paracentesis with albumin, TIPS, or transplant are used for tense or refractory ascites.

**44** **B** **Bacillus cereus**

Diarrhea · Step 1

B. cereus spores survive cooking. In rice left at room temperature they germinate and form a preformed, heat-stable emetic toxin (cereulide), causing vomiting within 1–6 hours. This is called "reheated rice syndrome".

WHY NOT THE OTHERS

(A) S. aureus also causes rapid vomiting from a preformed toxin, but the classic sources are mayonnaise, potato salad, and custards.

**45** **B** **CTR1, then ATP7A**

Physiology · Digestion and absorption · Step 1

Copper enters the enterocyte through CTR1 and reaches the blood through ATP7A.

**46** **A** **Esophageal manometry**

Esophageal motility disorders · Step 2

Esophageal manometry is the gold standard. In achalasia it shows incomplete LES relaxation, raised LES pressure, and absent distal peristalsis. Endoscopy is done first to exclude malignancy at the GE junction (pseudoachalasia).

**47** **E** **Repeat endoscopy every 3–5 years**

GERD · Step 2

Barrett esophagus without dysplasia is followed with endoscopic surveillance every 3–5 years to detect dysplasia. Dysplasia shortens the interval and may need endoscopic eradication therapy.

**48** **D** **Referred pain from nerves sharing a spinal level**

Localizing abdominal pain · Step 1

Blood from a ruptured spleen irritates the diaphragm, and the shoulder pain is referred because visceral and somatic nerves share a spinal level. Left shoulder pain with splenic rupture is the Kehr sign. He needs resuscitation and imaging.

WHY NOT THE OTHERS

(B) He did not injure the shoulder, and the timing follows the abdominal injury.

(C) Somatoparietal pain is felt at the site of the inflamed peritoneum.

(E) Visceral pain is dull and poorly localized to the abdomen, not felt at the shoulder.

## 49 (C) Endovascular angioplasty and stenting

Intestinal ischemia · Step 2

Symptomatic chronic mesenteric ischemia is treated with revascularization, usually endovascular angioplasty and stenting (or open bypass), plus risk factor control. It relieves pain and prevents progression to acute bowel infarction.

## 50 (D) False diverticula of mucosa and submucosa herniating at vasa recta entry points

Diverticular diseases · Step 1

Colonic diverticula are false (pulsion) diverticula. Mucosa and submucosa herniate through weak points in the muscularis externa where the vasa recta enter. High intraluminal pressure in the sigmoid, from low fiber, constipation, and aging, drives their formation.

### WHY NOT THE OTHERS

(C) An outpouching above the cricopharyngeus is a Zenker diverticulum.

(E) A vitelline duct remnant is a Meckel diverticulum, which is a true diverticulum.